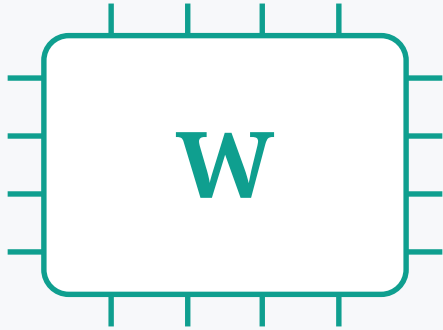




# Liveness Proofs

## for Hardware Model Checking

CAV 2026

Nils Froleys

KU LEUVEN

Emily Yu



Universiteit  
Leiden

Bart Bogaerts

KU LEUVEN

Armin Biere

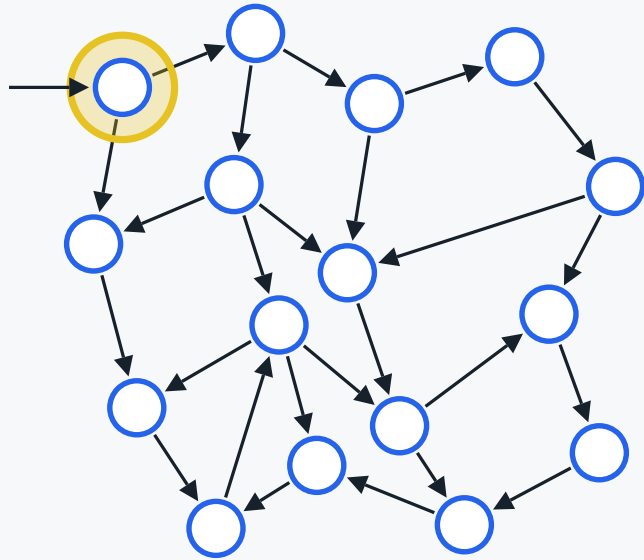
universität freiburg

Keijo Heljanko

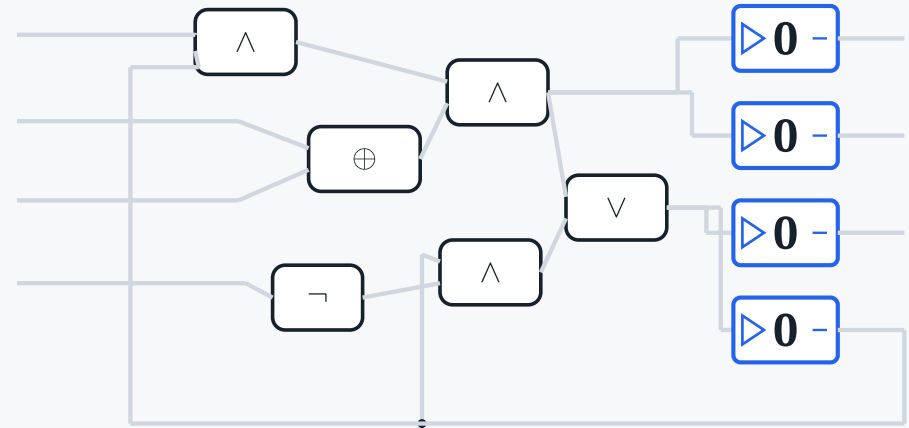


UNIVERSITY OF HELSINKI

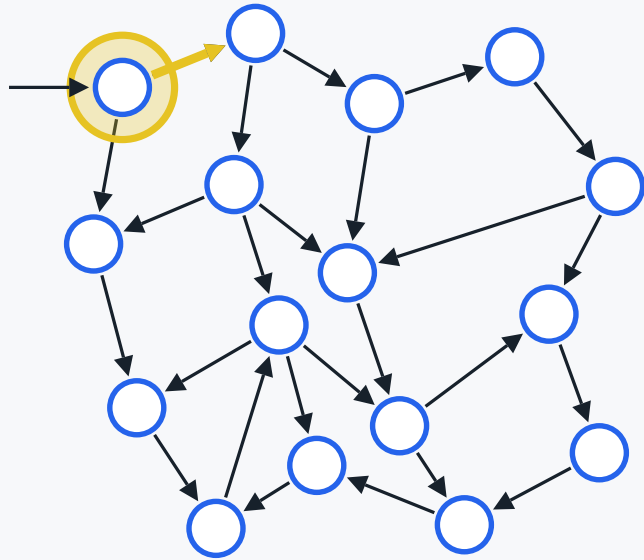
# Boolean Circuits



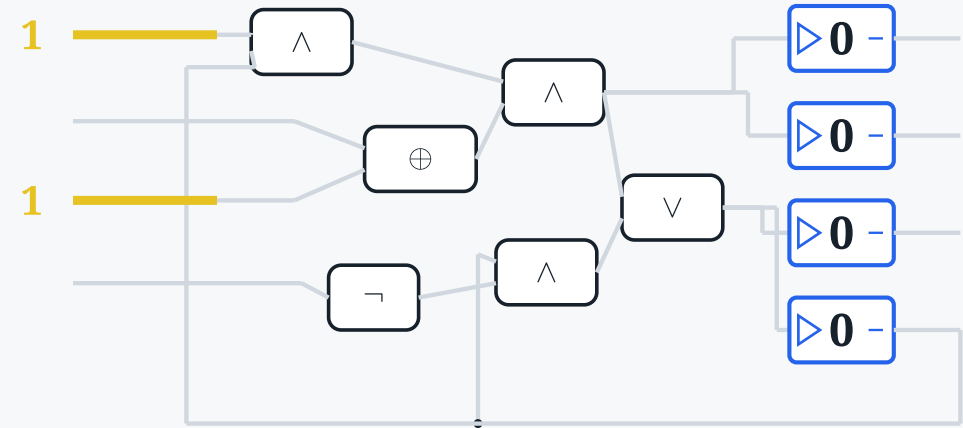
$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$



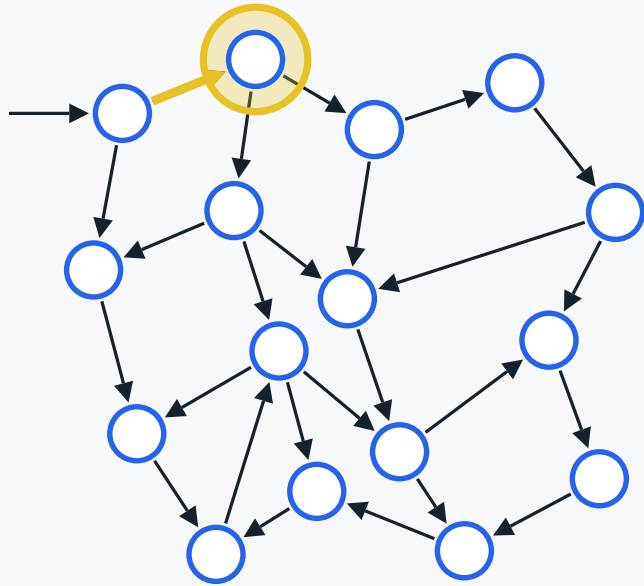
# Boolean Circuits



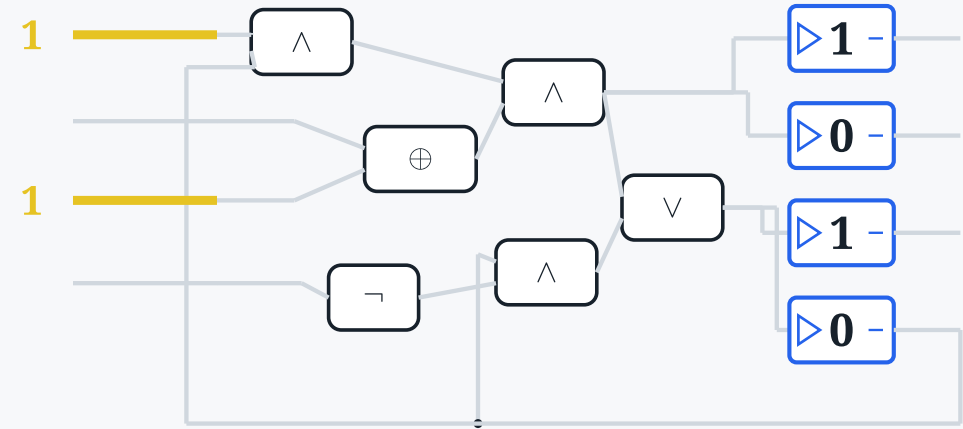
$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$



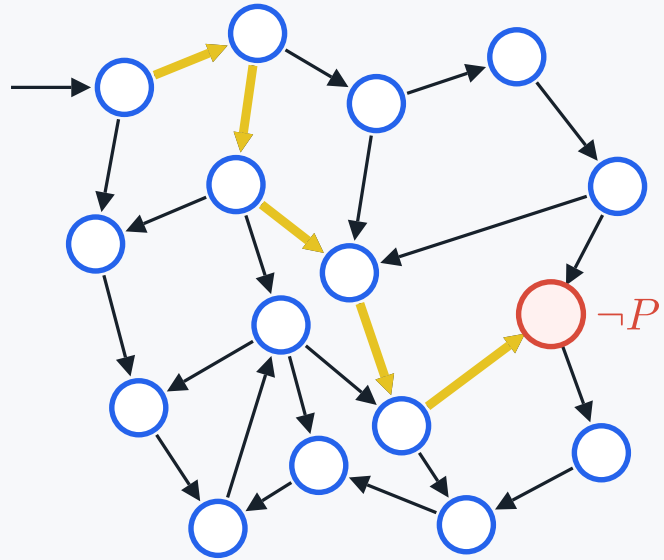
# Boolean Circuits



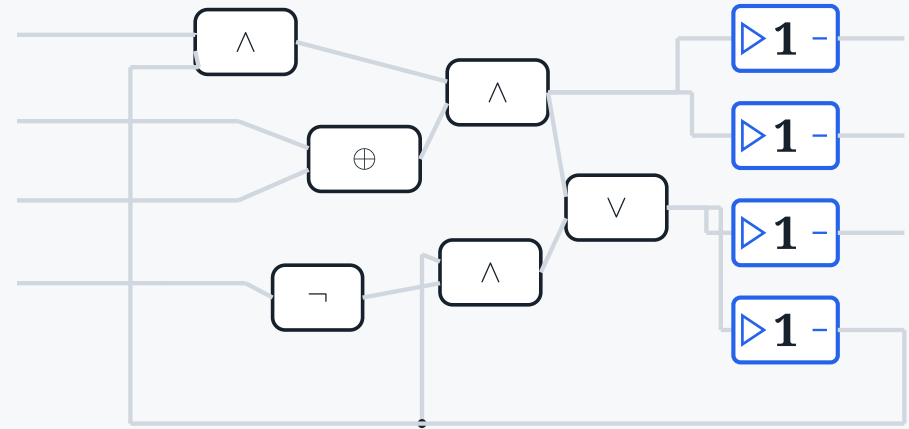
$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$



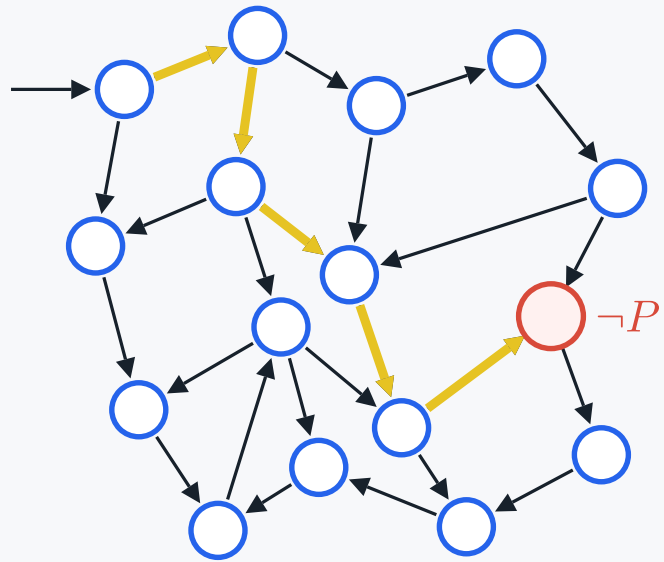
# Boolean Circuits



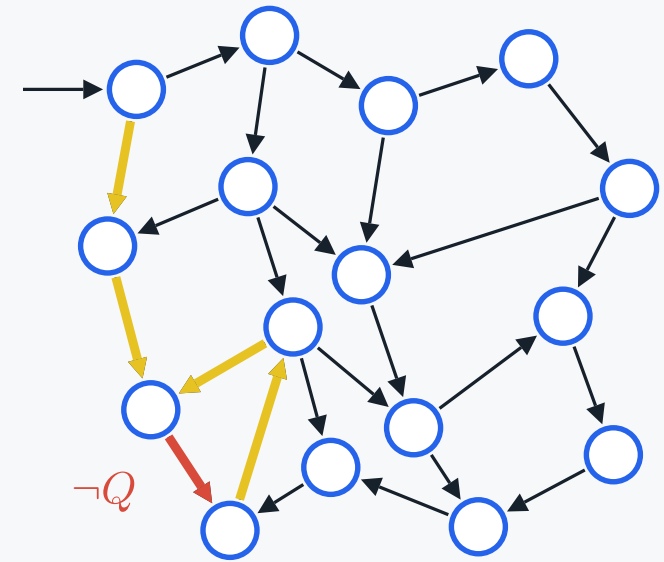
$$\mathcal{M} = \langle \mathcal{I}, \mathcal{L}, \mathcal{R}, \mathcal{F}, \mathcal{C}, \mathcal{P}, \mathcal{Q} \rangle$$



# Boolean Circuits

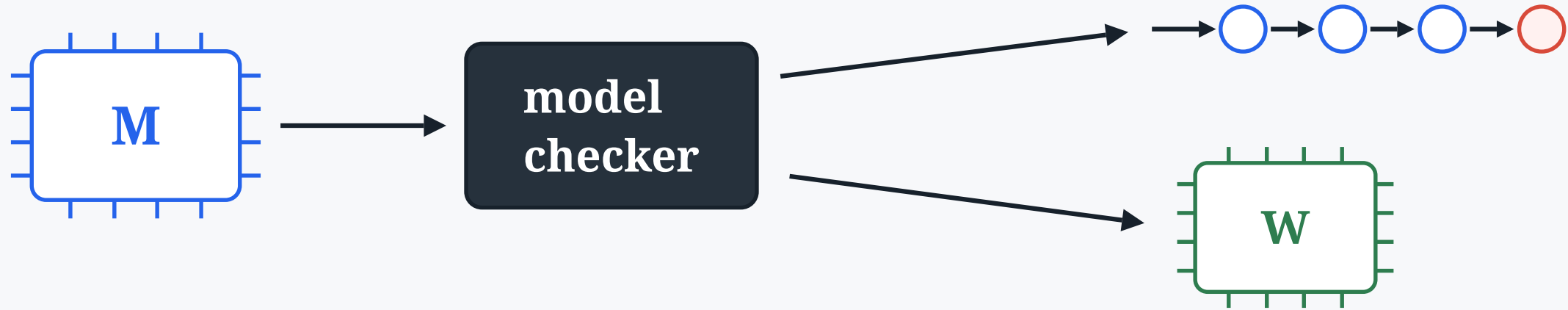


**Safety**

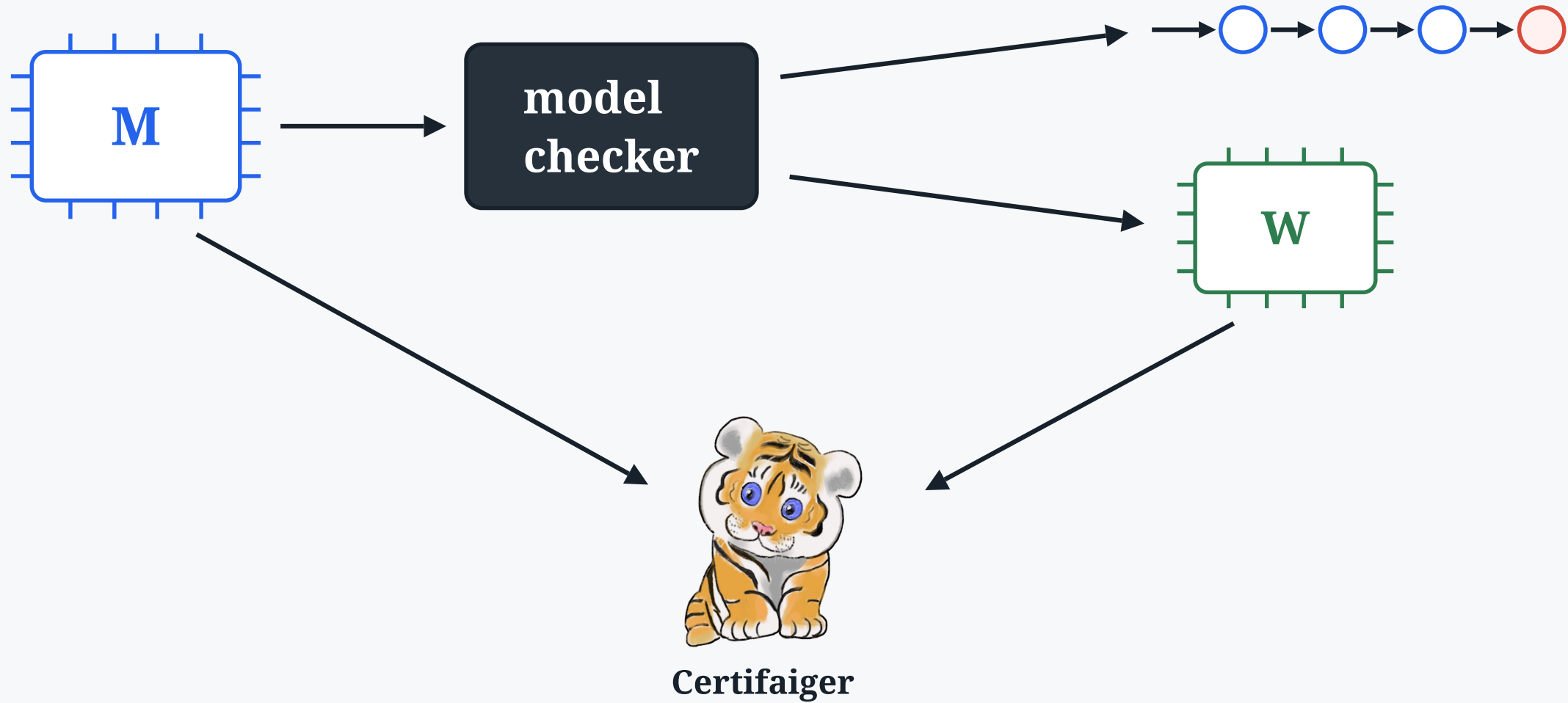


**Liveness**

# Certificates



# Certificates



# Safety

---

- [CAV21] IC3, BDD, Interpolation
- [FMCAD22] k-Induction
- [FMCAD23] Temporal Decomposition
- [IJCAR24] Cone of Influence, Phase Abstraction
- [CAV25] Case Study: Certifying HWMCC
- [FM26] Uniqueness Constraints, Constraint Mining
- [IJCAR26] Word-Level

# Liveness

---

# Safety

---

- | [CAV21] IC3, BDD, Interpolation
- | [FMCAD22] k-Induction
- | [FMCAD23] Temporal Decomposition
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# Liveness

---

- | **k-Liveness**
- | **Stabilizer Extraction**
- | **Liveness to Safety**
- | **rLive**

# Safety

---

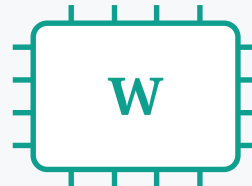
- [CAV21] IC3, BDD, Interpolation
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- [IJCAR24] Cone of Influence, Phase Abstraction
- [CAV25] Case Study: Certifying HWMCC
- [FM26] Uniqueness Constraints, Constraint Mining
- [IJCAR26] Word-Level

# Liveness

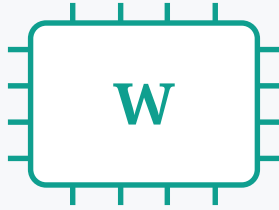
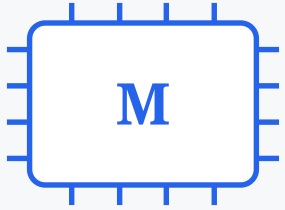
---

- k-Liveness
- Stabilizer Extraction
- Liveness to Safety
- rLive

one generic certificate



# Certificate Checking



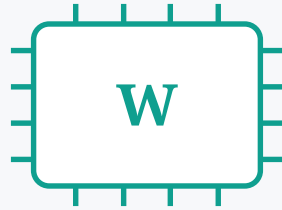
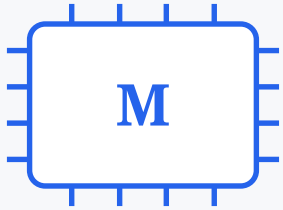
**Simulates**



**Inductive**

**Ranked**

# Certificate Checking



**Simulates**

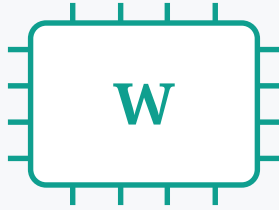
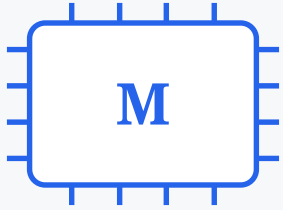
$W$  is safe and live  $\Rightarrow M$  is safe and live



**Inductive**

**Ranked**

# Certificate Checking



**Simulates**

$W$  is safe and live  $\Rightarrow M$  is safe and live



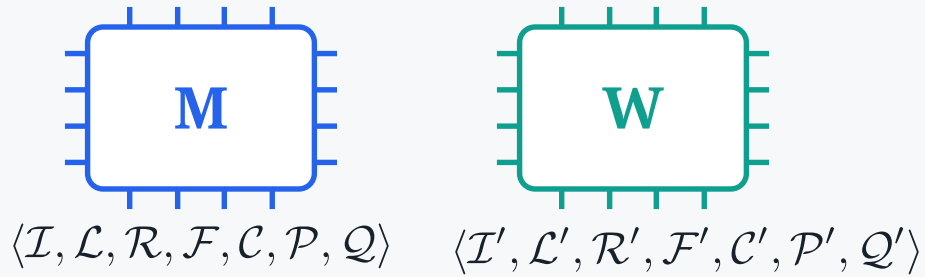
**Inductive**

$W$  is safe

**Ranked**

$W$  is live

# Certificate Checking



Simulates



Inductive

Ranked

**Reset**

$$\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$$

**Transition**

$$F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$$

**Safety**

$$C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$$

**Liveness**

$$\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$$

**Base**

$$\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$$

**Induction**

$$F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$$

**Decrease**

$$\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$$

**Closure**

$$\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$$

8 SAT Checks

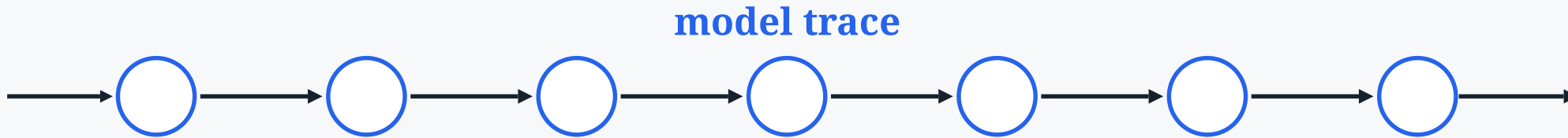
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



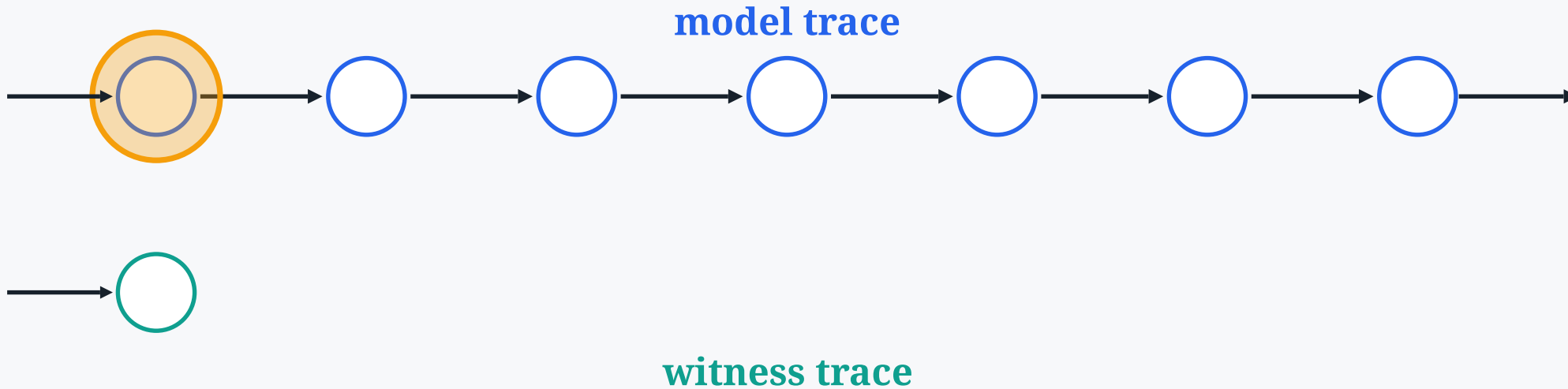
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



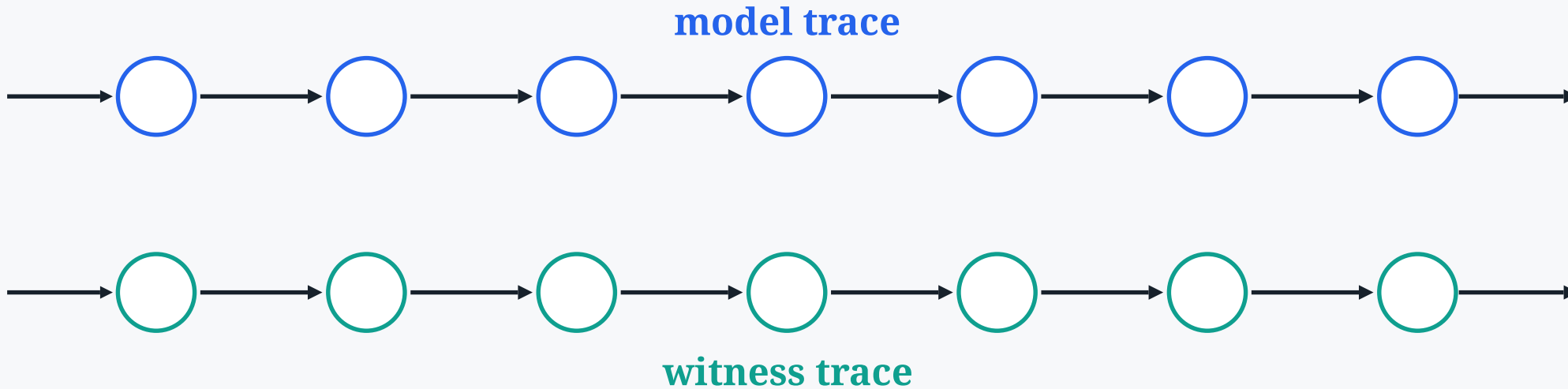
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

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**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



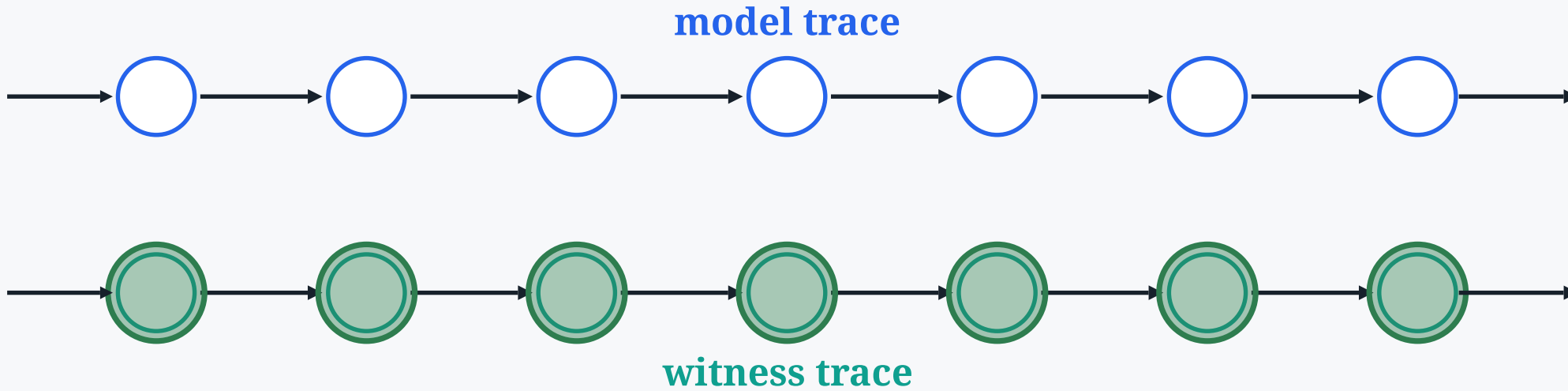
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



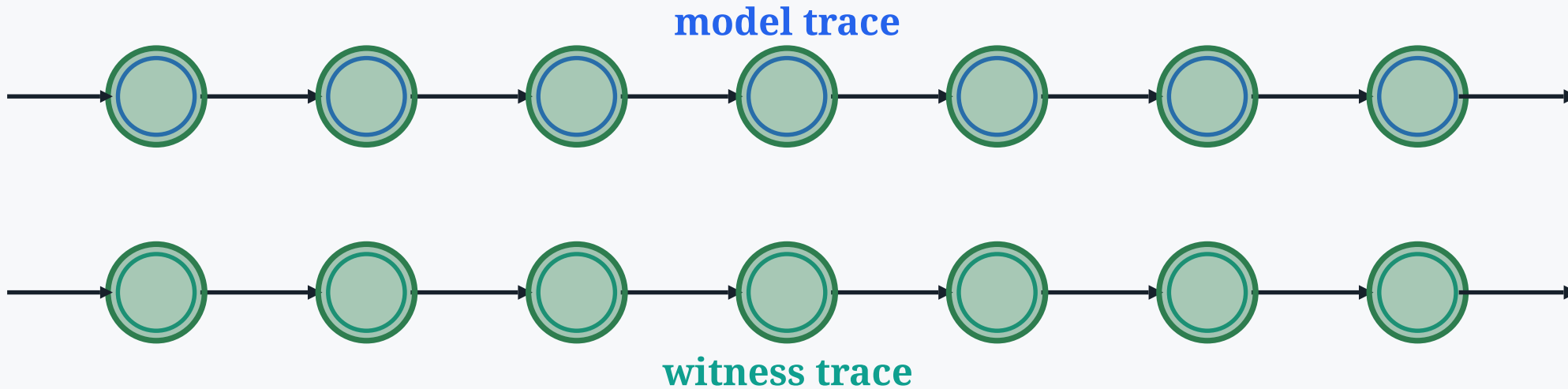
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

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**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



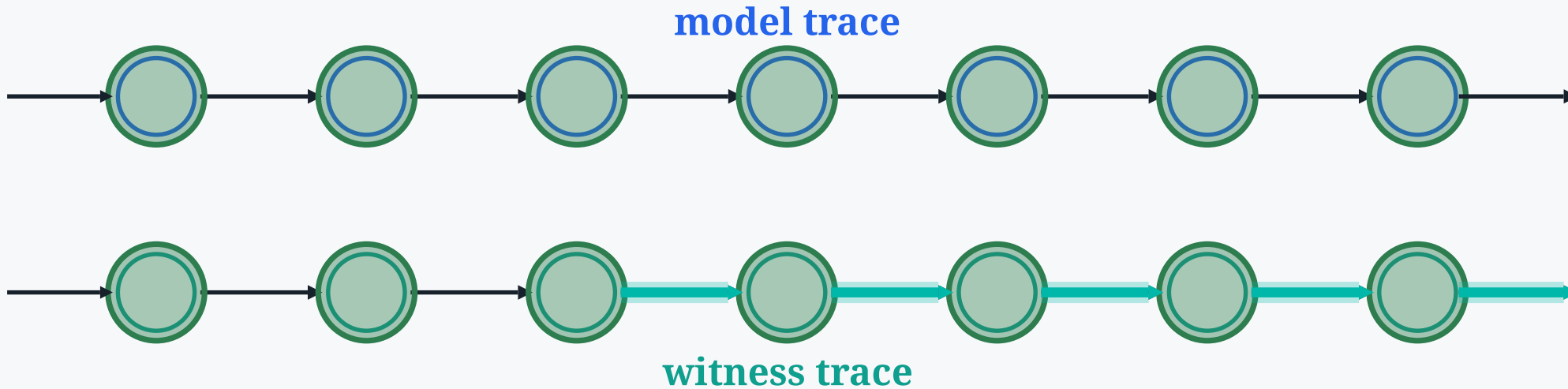
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



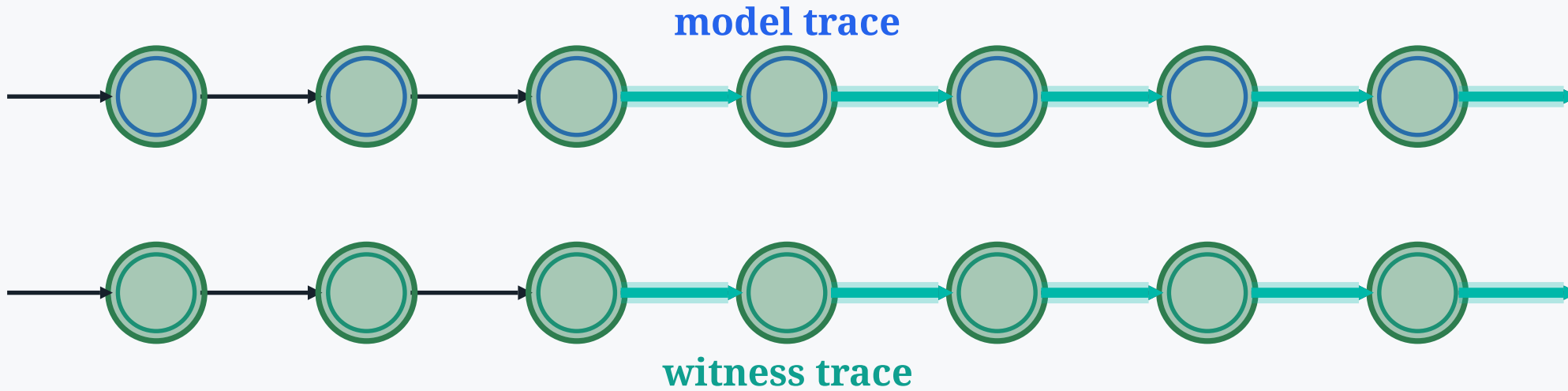
# Simulation

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

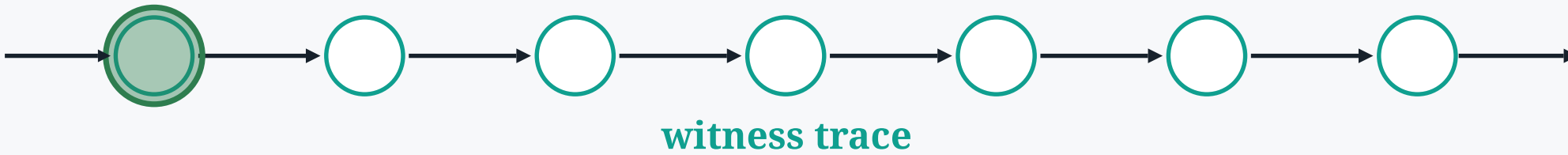
**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$



# Inductive

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$



# Inductive

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

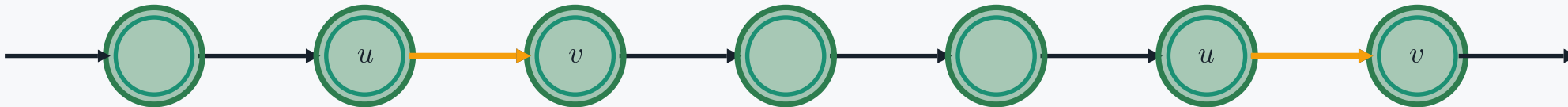
**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$



# Ranked

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

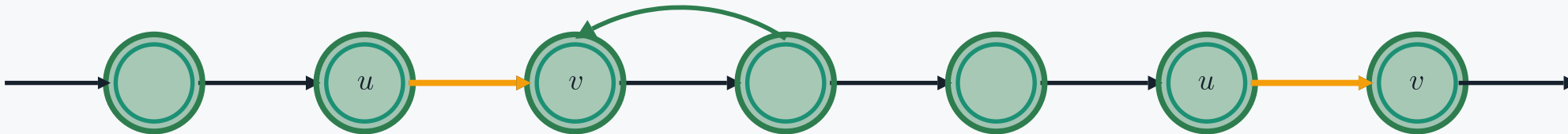
**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



# Ranked

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

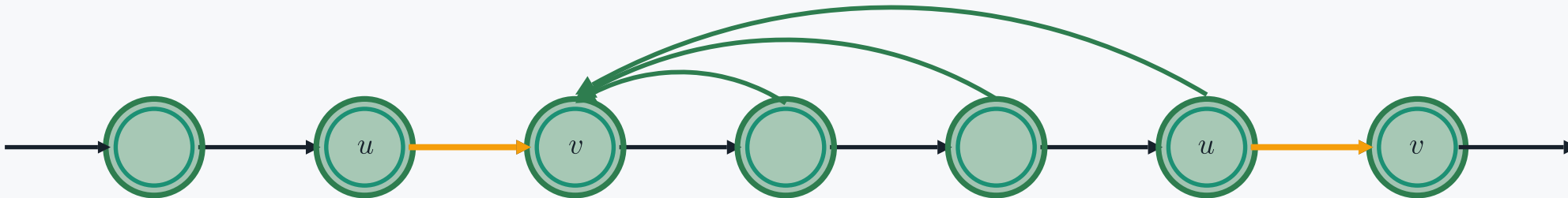
**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



# Ranked

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

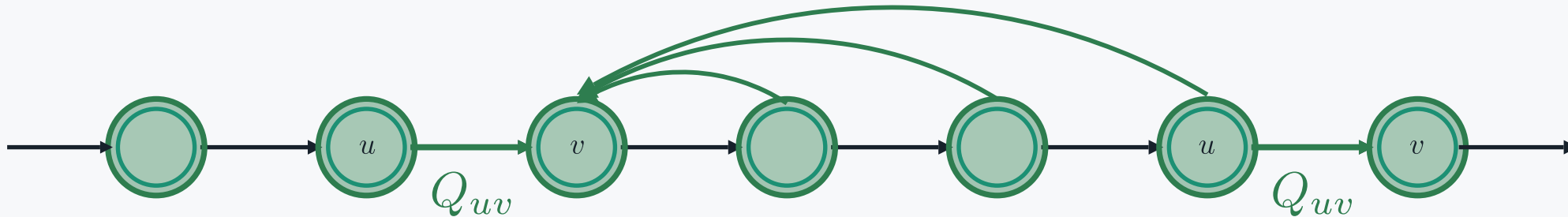
**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



# Ranked

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



$Q$  holds on repeated edges

# Safety

---

- | [CAV21] IC3, BDD, Interpolation
- | [FMCAD22] k-Induction
- | [FMCAD23] Temporal Decomposition
- | [IJCAR24] Cone of Influence, Phase Abstraction
- | [CAV25] Case Study: Certifying HWMCC
- | [FM26] Uniqueness Constraints, Constraint Mining
- | [IJCAR26] Word-Level

# Liveness

---

- | **k-Liveness**
- | **Stabilizer Extraction**
- | **Liveness to Safety**
- | **rLive**

# k-Liveness

Claessen & Sörensson [FMCAD12]



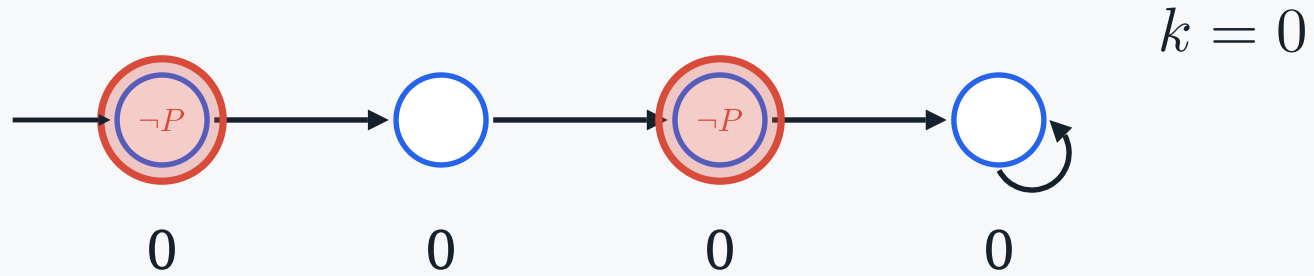
# k-Liveness

Claessen & Sörensson [FMCAD12]



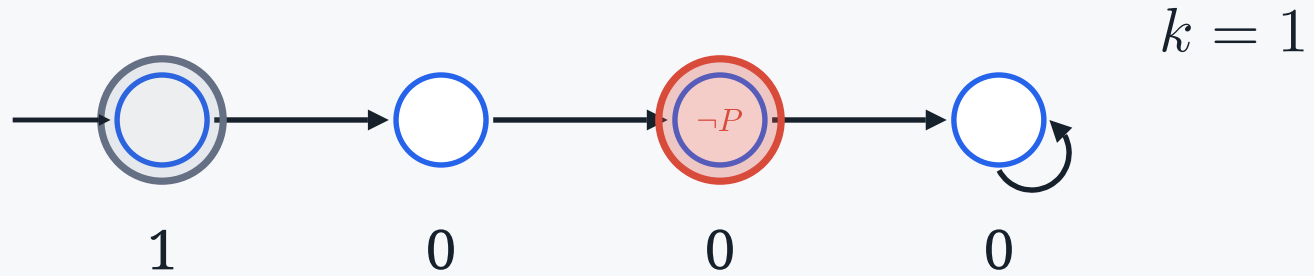
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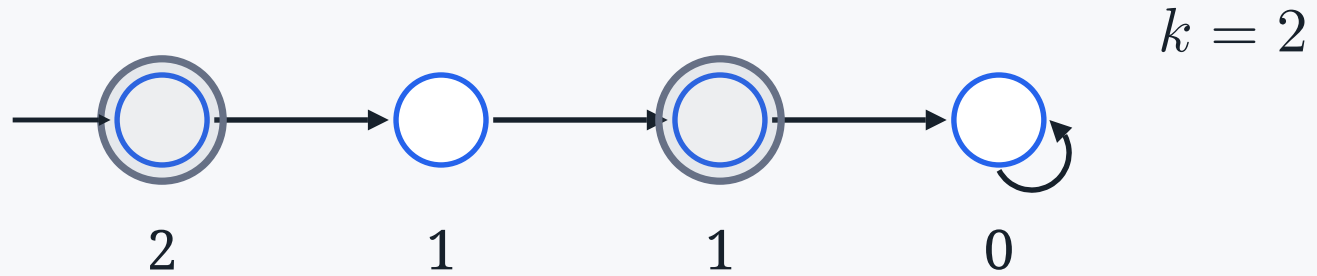
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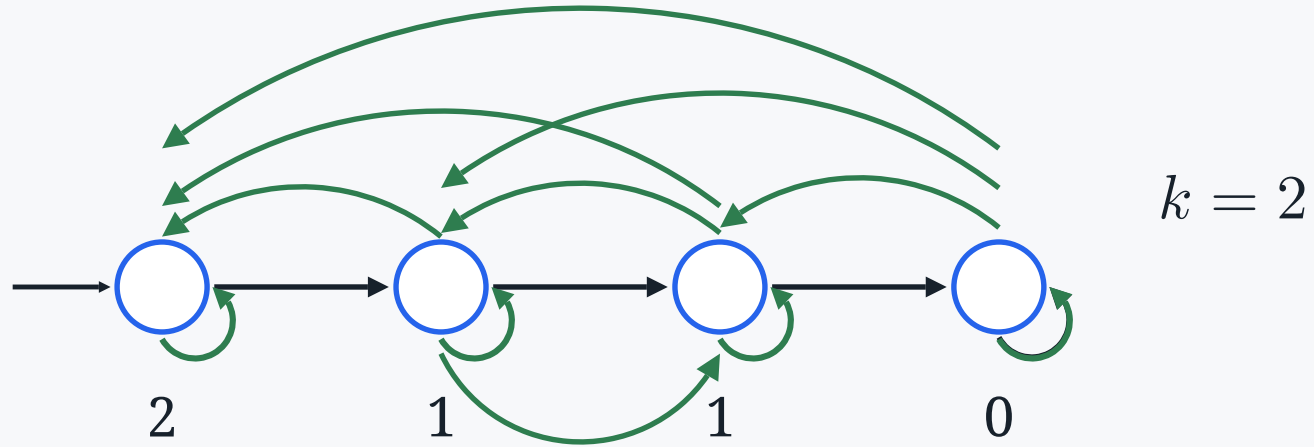
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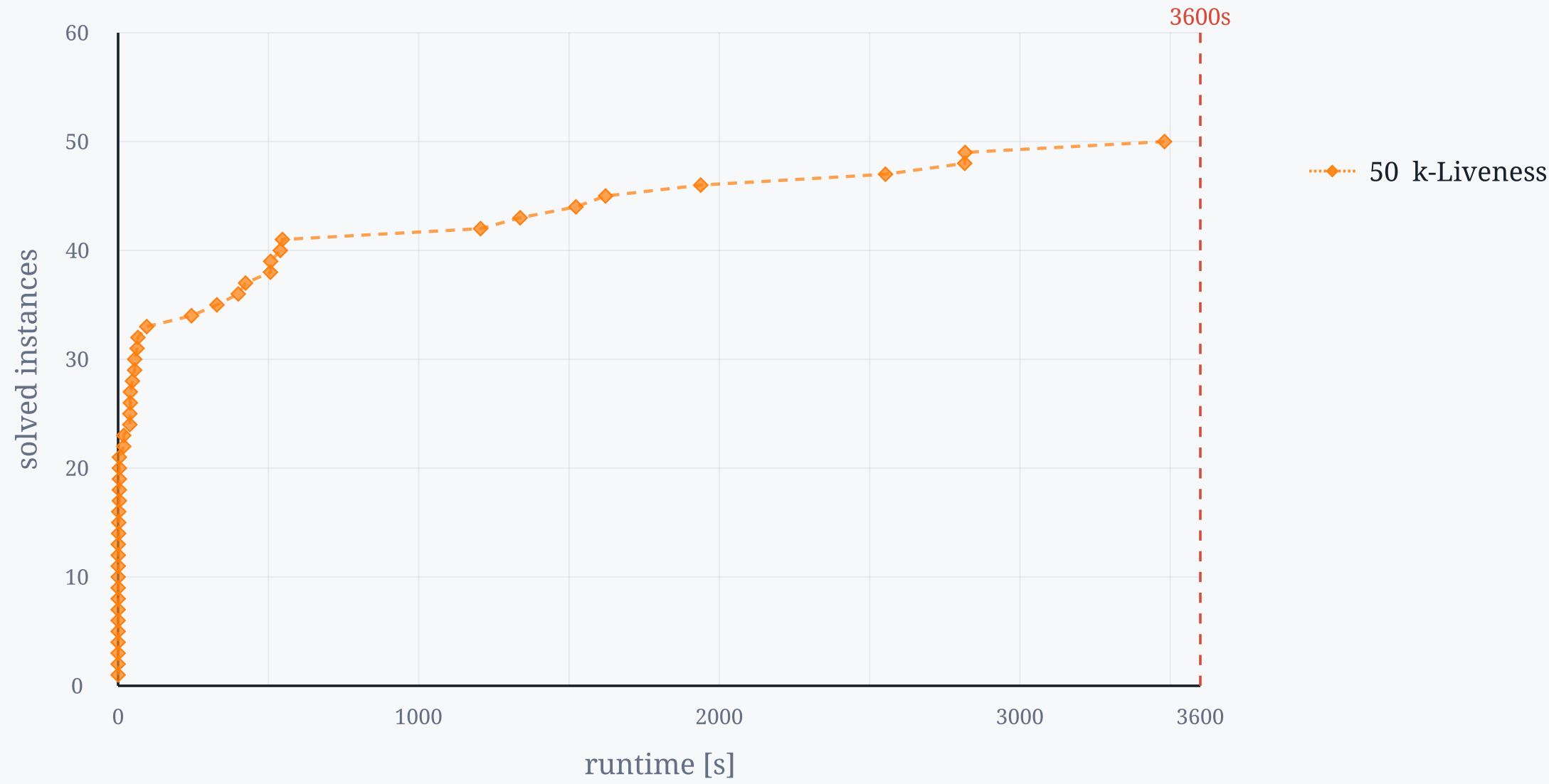


# k-Liveness

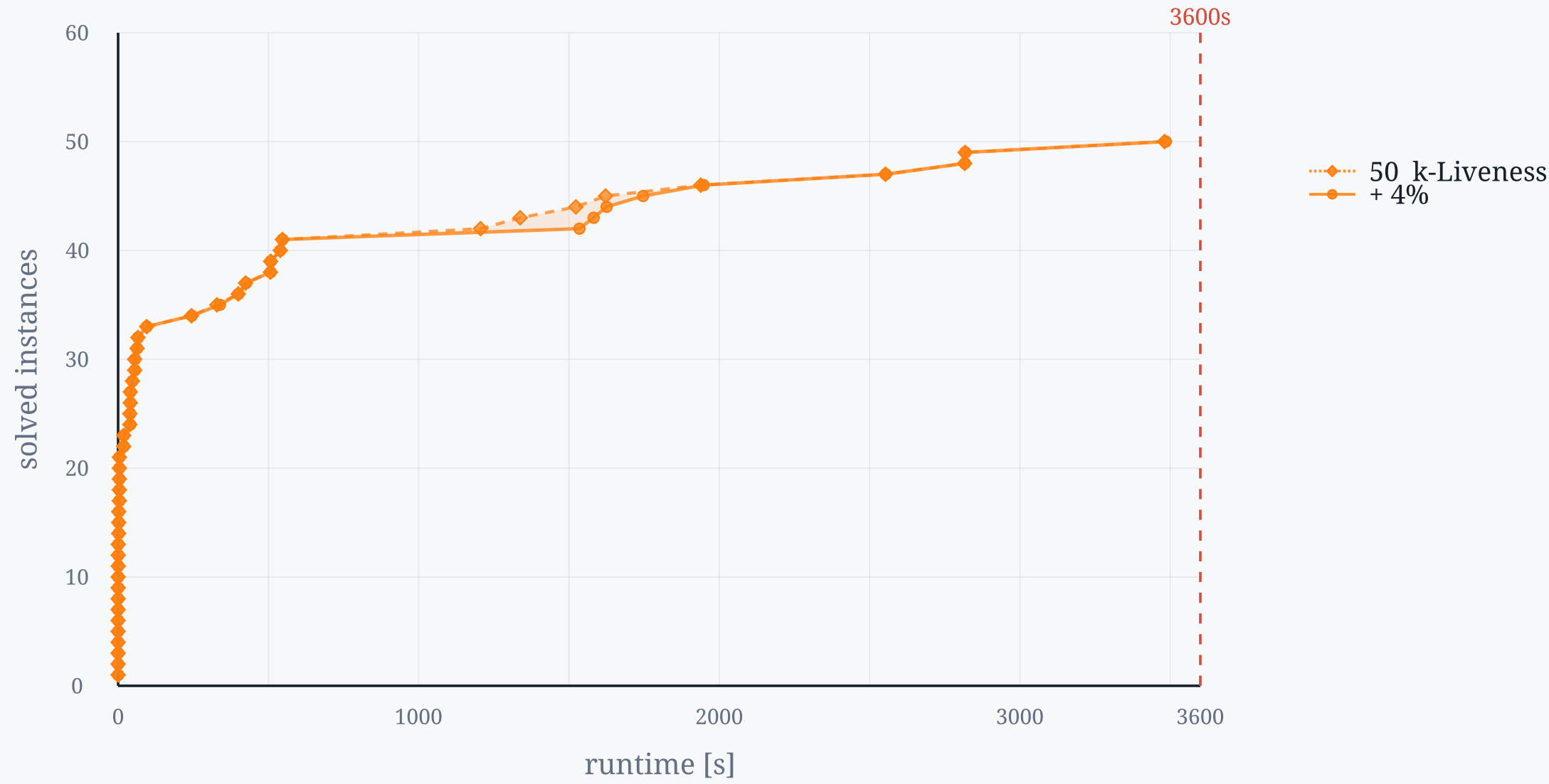
Claessen & Sörensson [FMCAD12]



# HWMCC25

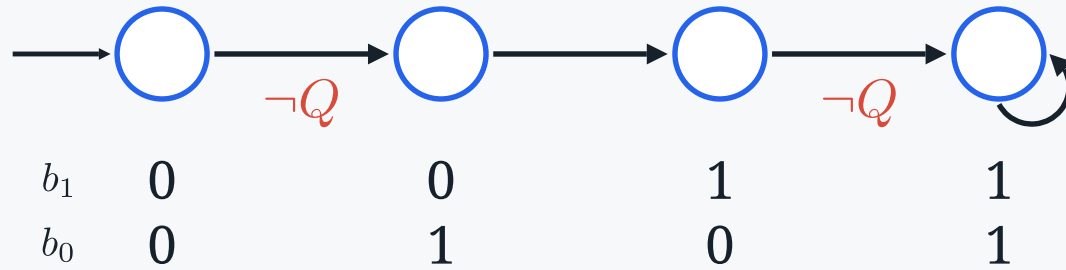


# HWMCC25



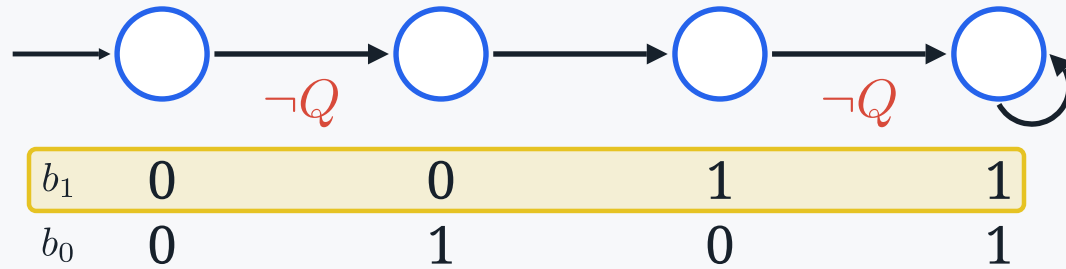
# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



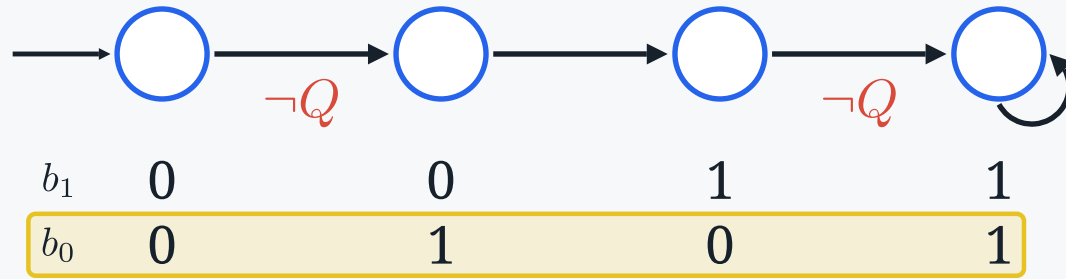
# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

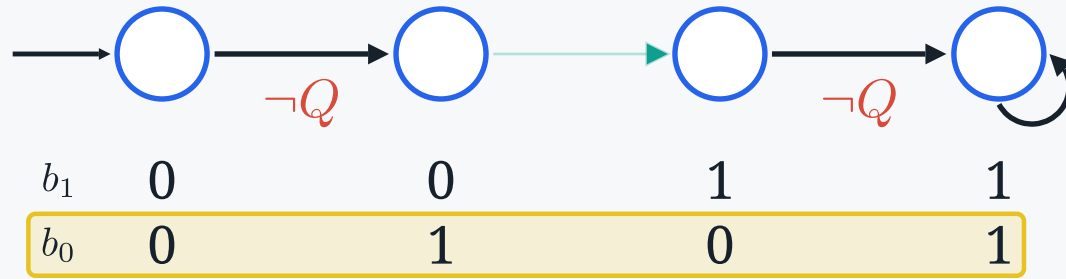
# Stabilizer Extraction



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

# Stabilizer Extraction

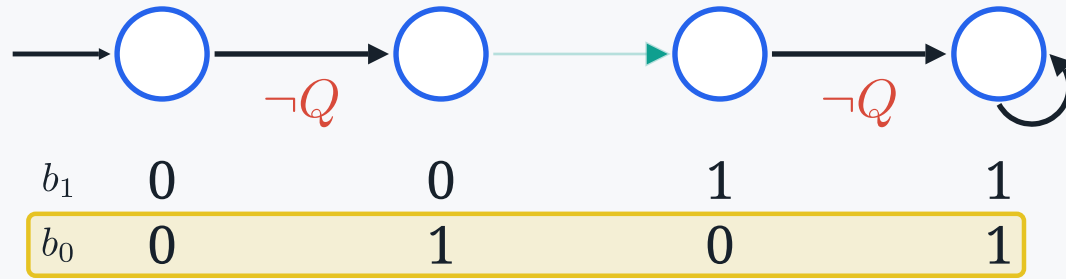


$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



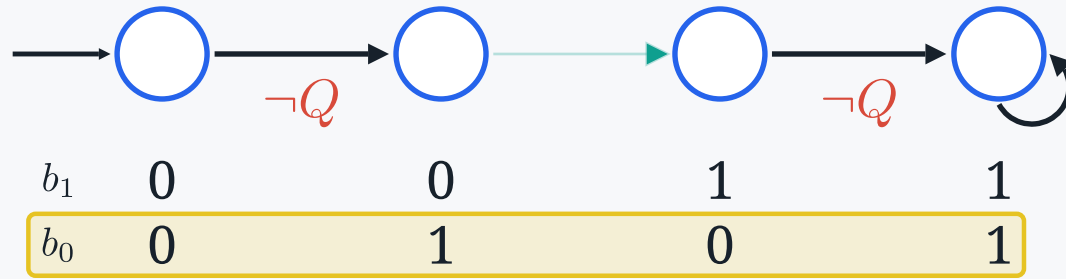
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$



# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



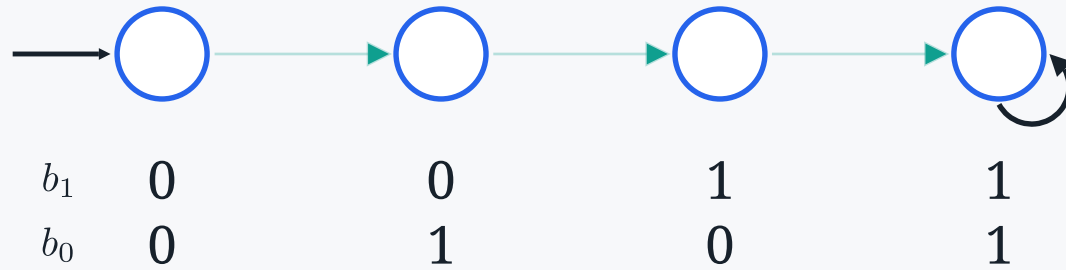
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$



# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



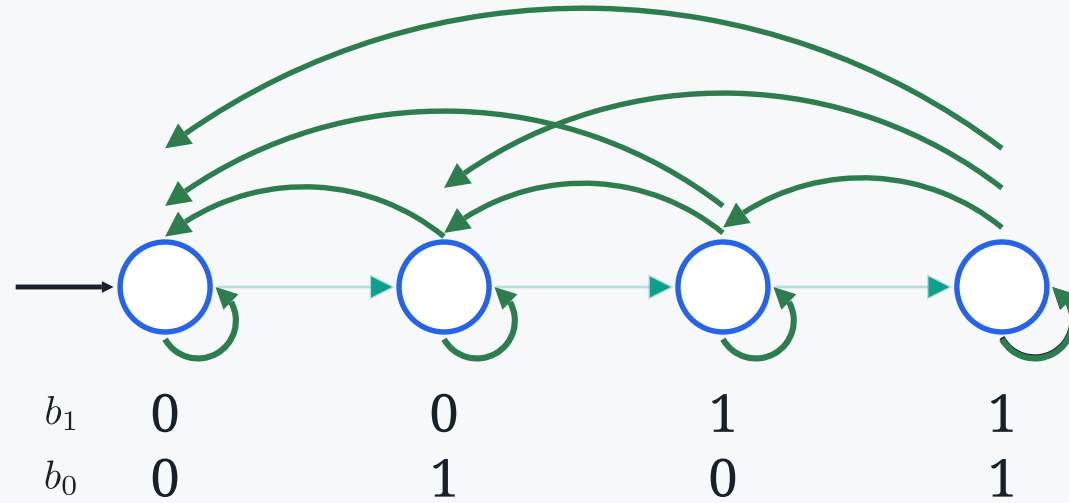
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

$$(b_s^1 \Leftrightarrow b_t^1) \wedge (b_s^0 \Leftrightarrow b_t^0) \Rightarrow Q$$

# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



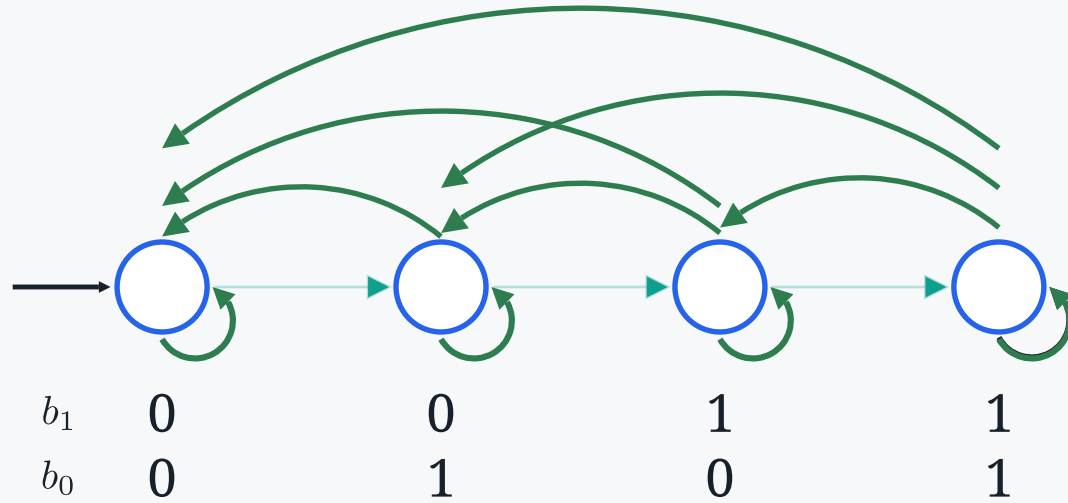
$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

$$(b_s^1 \Leftrightarrow b_t^1) \wedge (b_s^0 \Leftrightarrow b_t^0) \Rightarrow Q$$

# Stabilizer Extraction

Claessen & Sörensson [FMCAD12]



$$F_{st} \Rightarrow (b_s^1 \Rightarrow b_t^1)$$

$$F_{st} \wedge (b_s^1 \Leftrightarrow b_t^1) \Rightarrow (b_s^0 \Rightarrow b_t^0)$$

$$(b_s^1 \Leftrightarrow b_t^1) \wedge (b_s^0 \Leftrightarrow b_t^0) \Rightarrow Q$$

$$Q' := \underbrace{\text{dec}_n}_{\text{less stable}} \vee \left( \underbrace{\text{eq}_n}_{\text{equally stable}} \wedge Q^{\text{k-liveness}} \right)$$

# HWMCC25



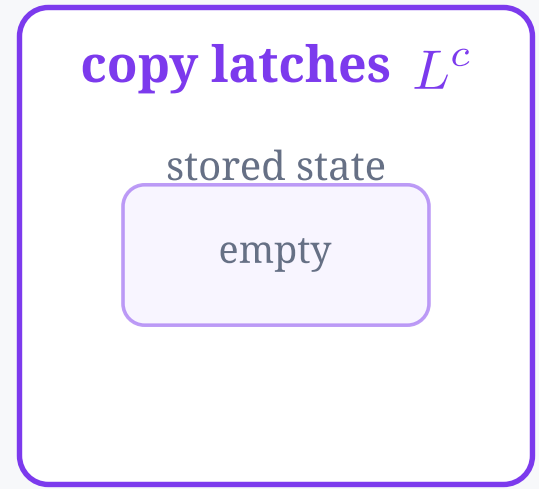
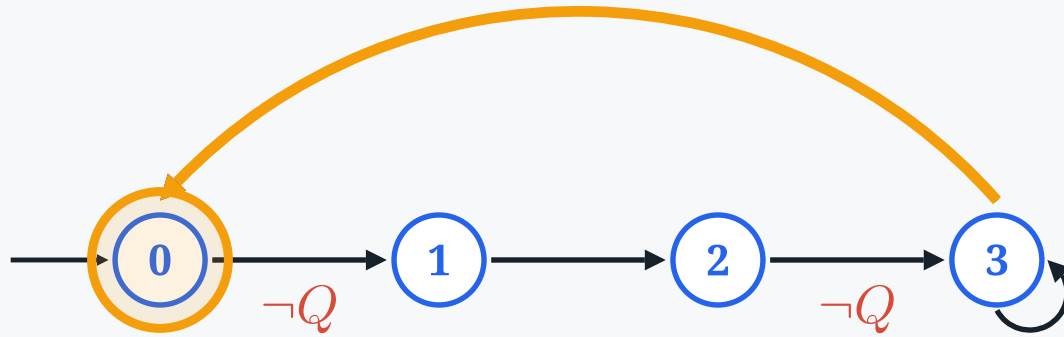
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



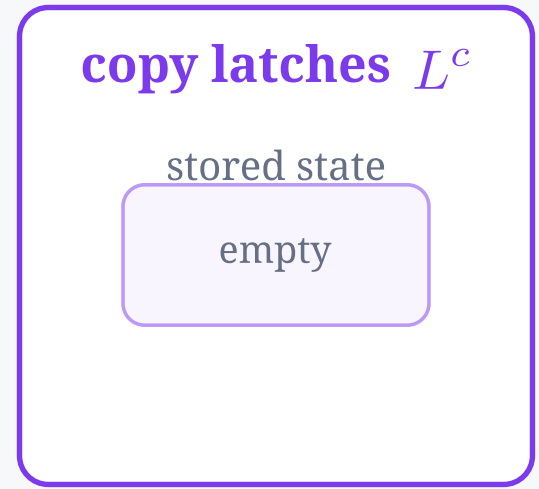
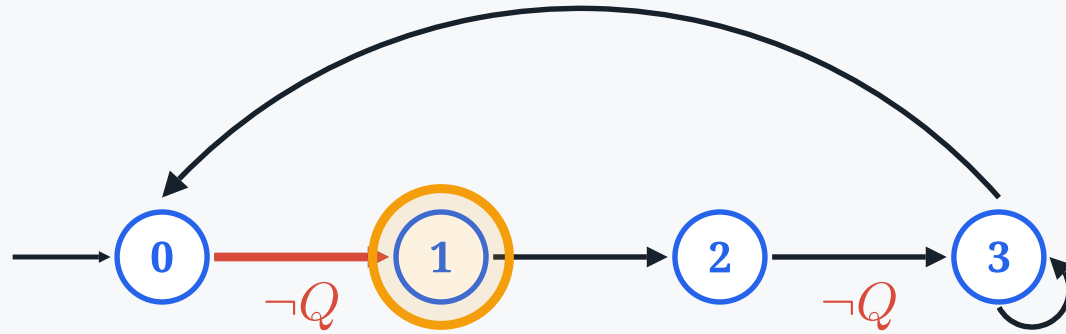
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



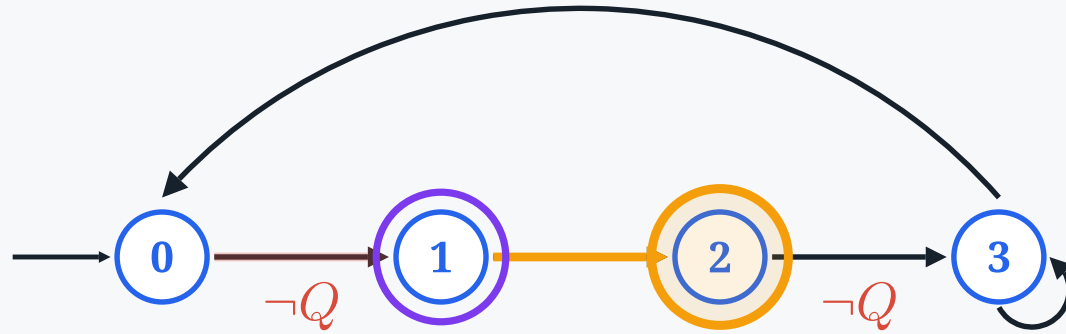
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



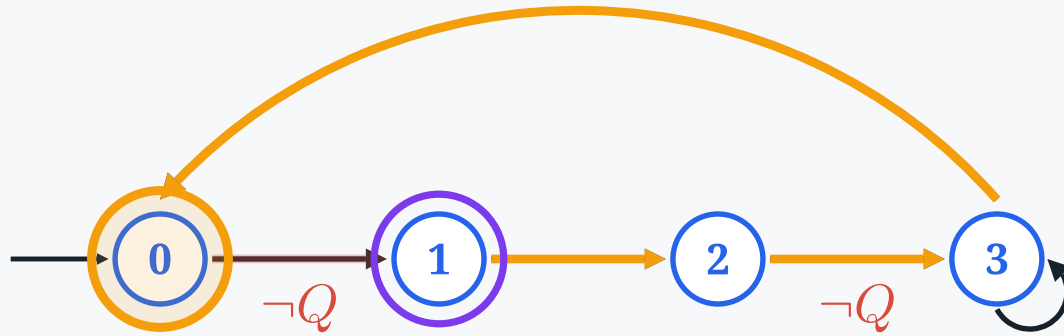
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



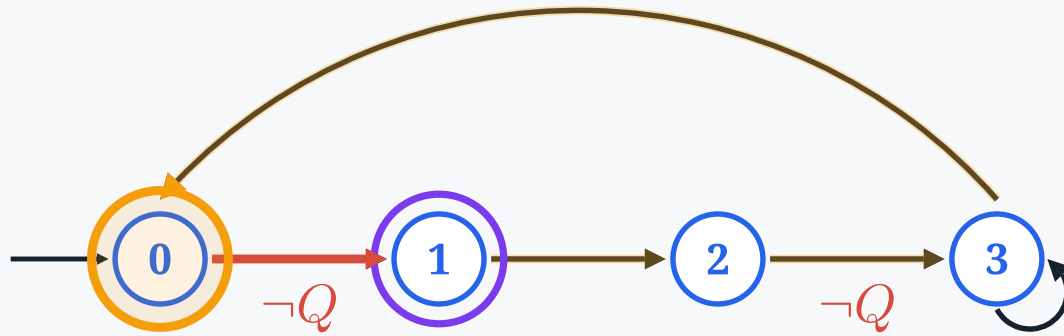
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



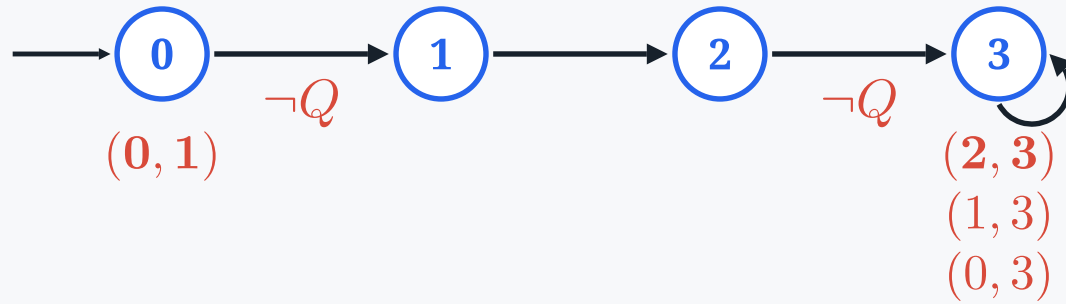
# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]



# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]

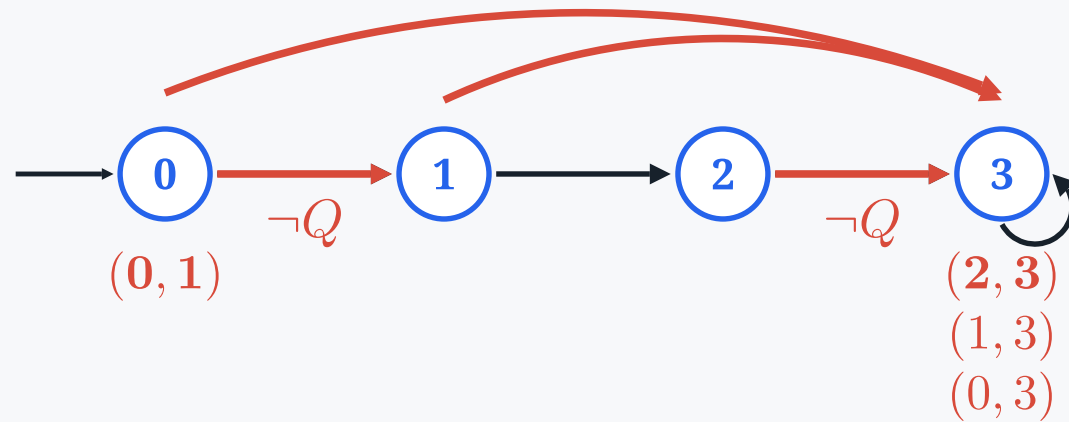


$(L, L^c)$   
current state stored state

$$P_{st}^{L2S} : F_{st} \wedge (L_t = L^c) \Rightarrow Q_{st}$$

# Liveness to Safety

Biere, Artho & Schuppan [FMICS02]

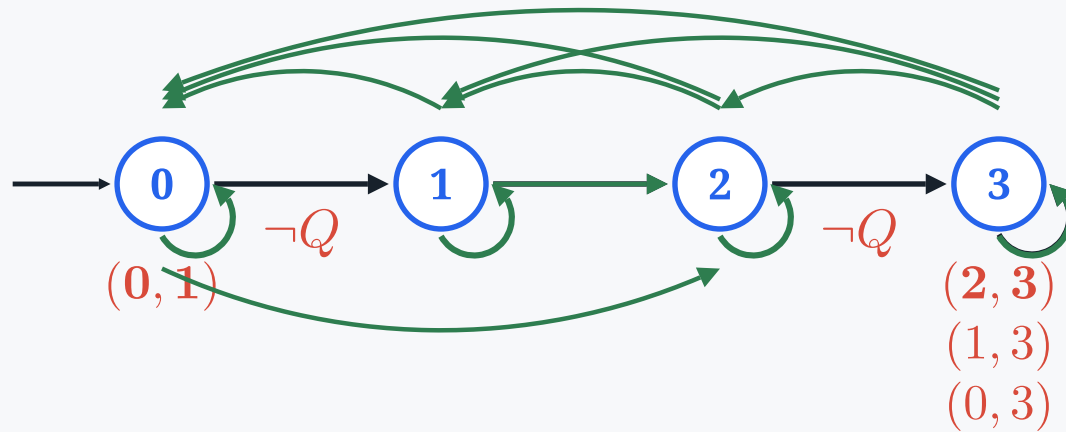


$(L, L^c)$   
current state stored state

$$P_{st}^{L2S} : F_{st} \wedge (L_t = L^c) \Rightarrow Q_{st}$$

# Liveness to Safety

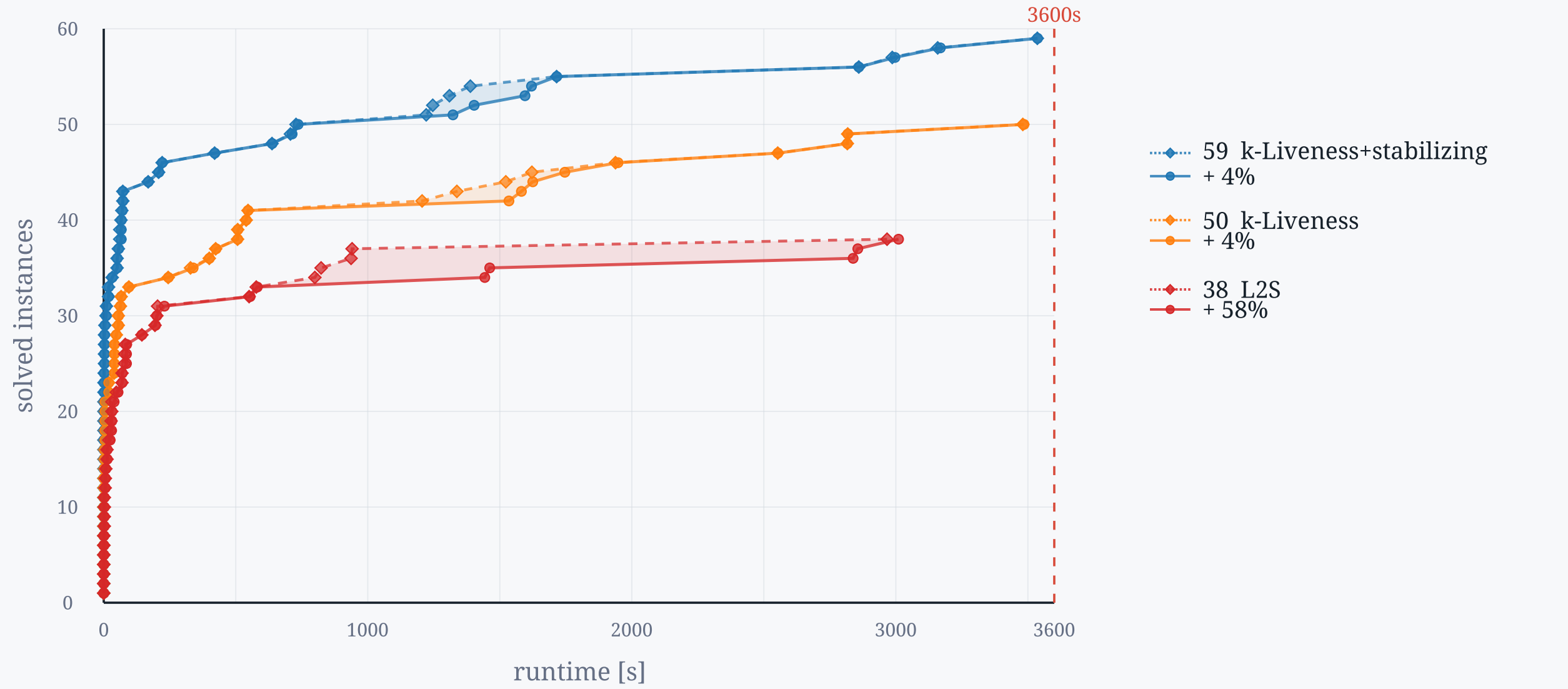
Biere, Artho & Schuppan [FMICS02]



$(L, L^c)$   
 current state    stored state

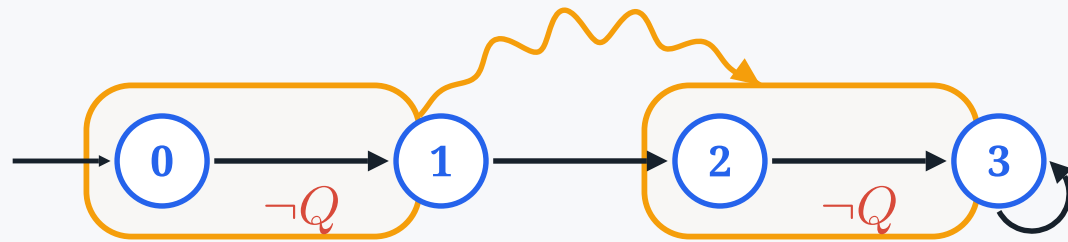
$$P_{st}^{L2S} : F_{st} \wedge (L_t = L^c) \Rightarrow Q_{st}$$

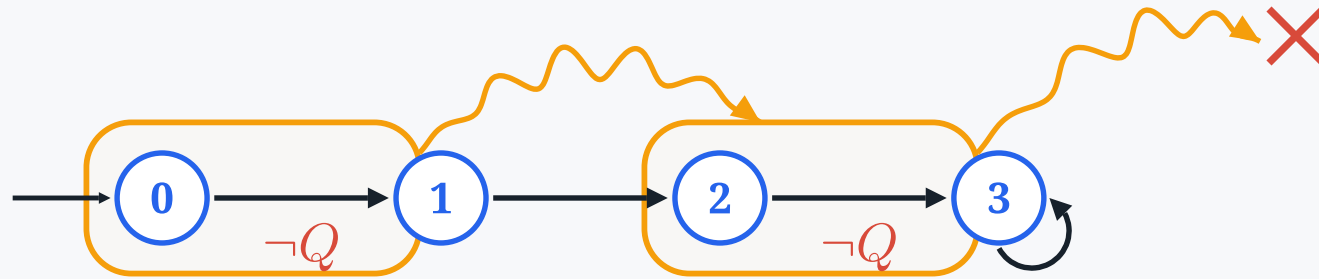
# HWMCC25

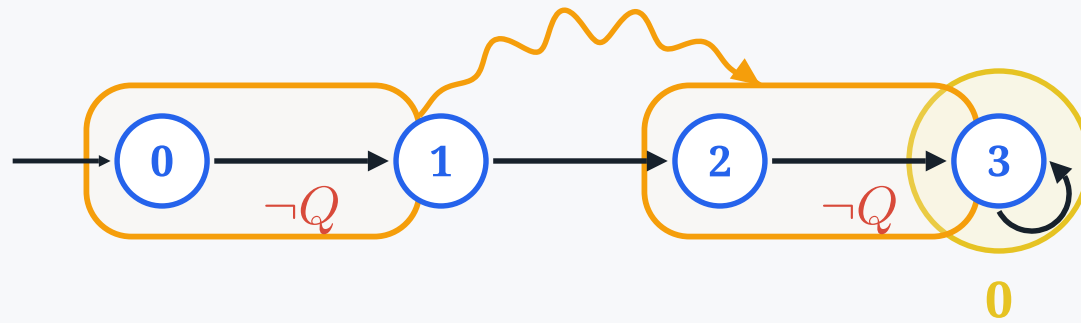


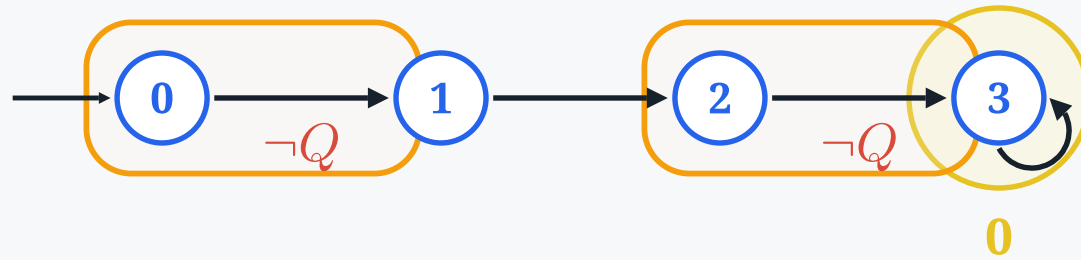


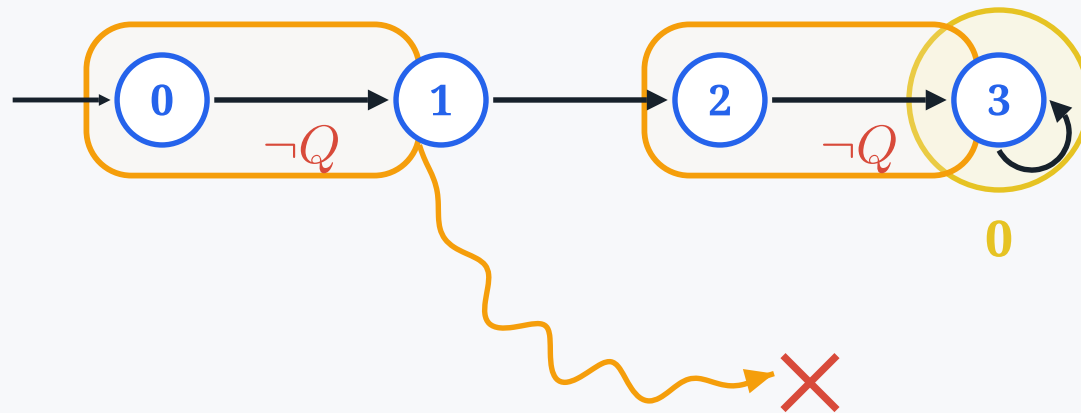


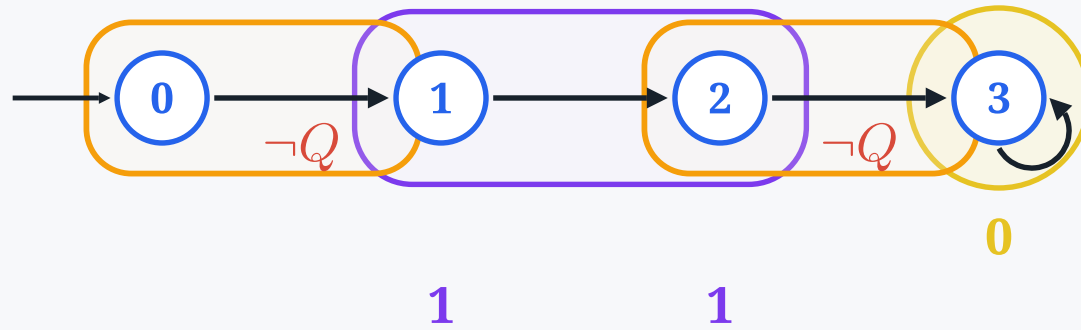


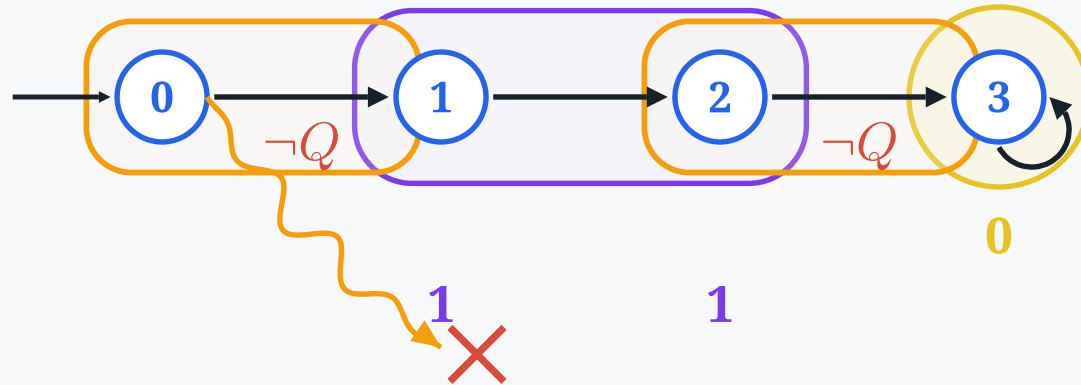


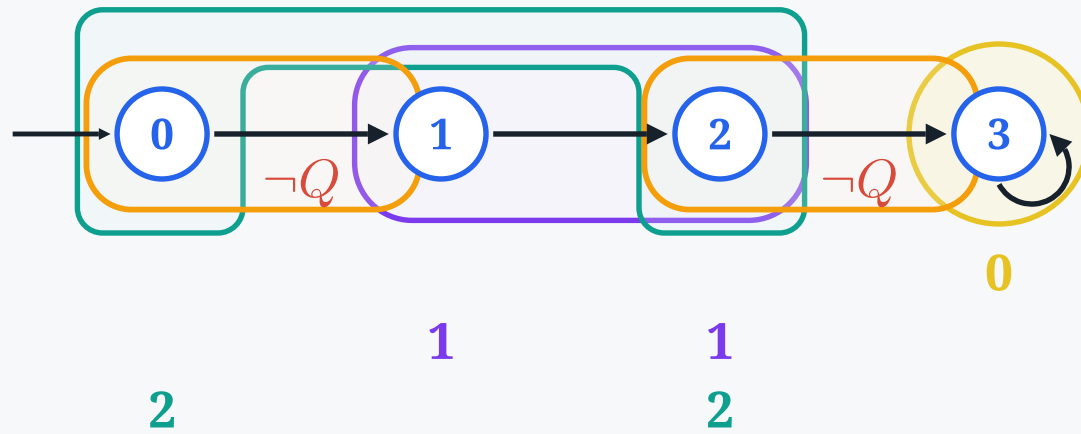


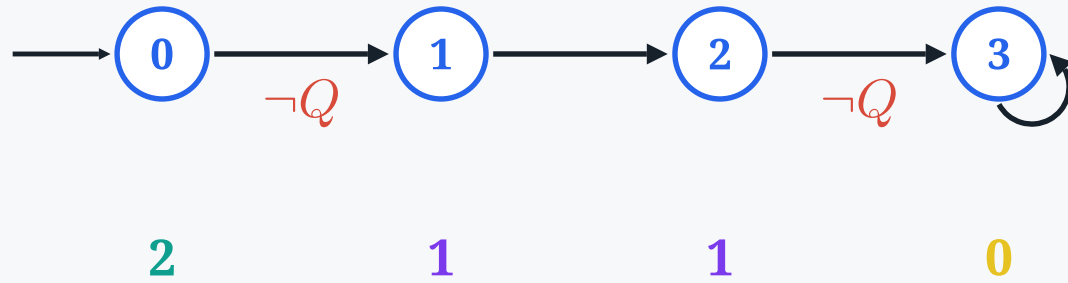


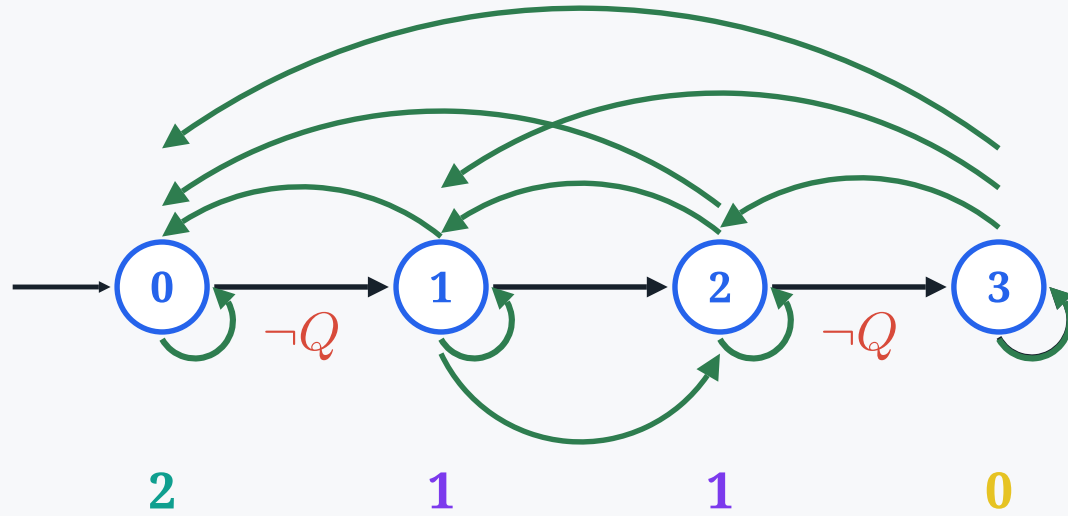




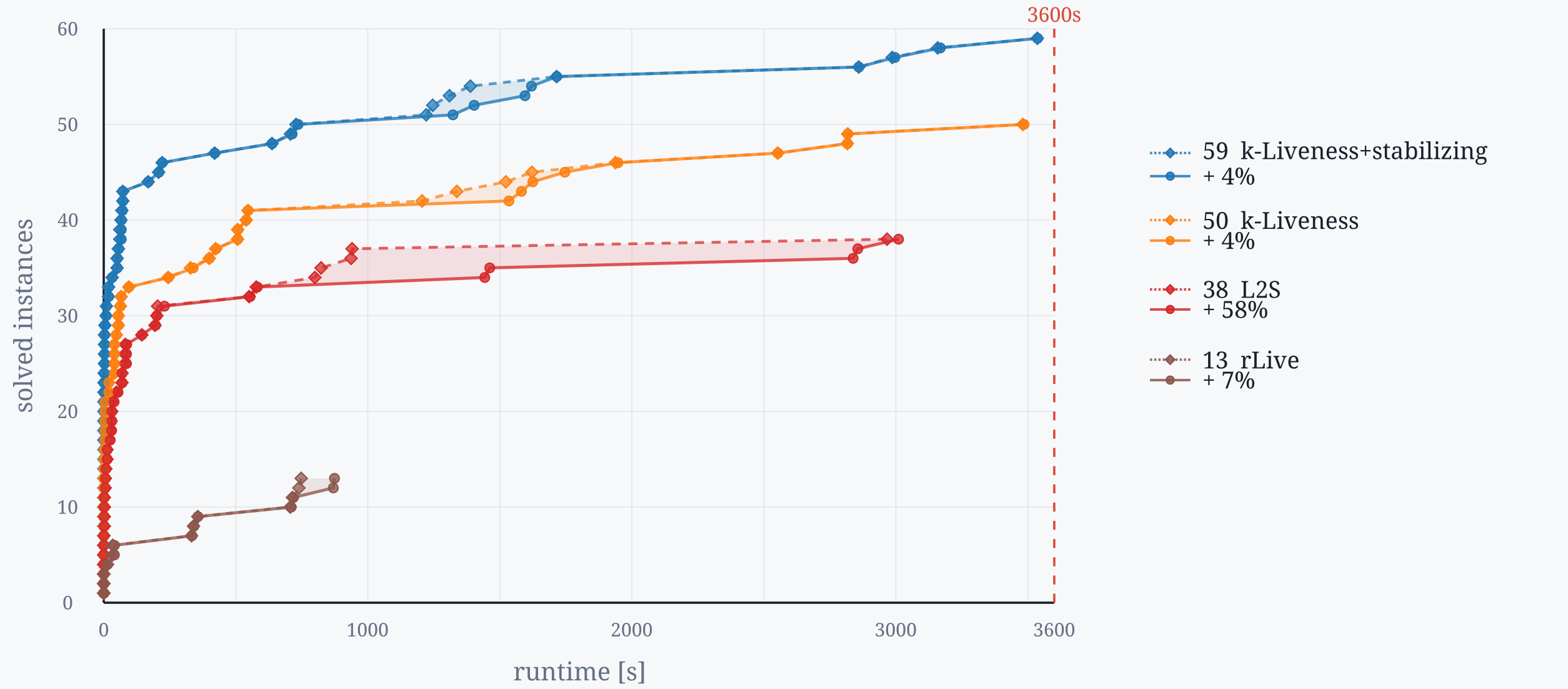




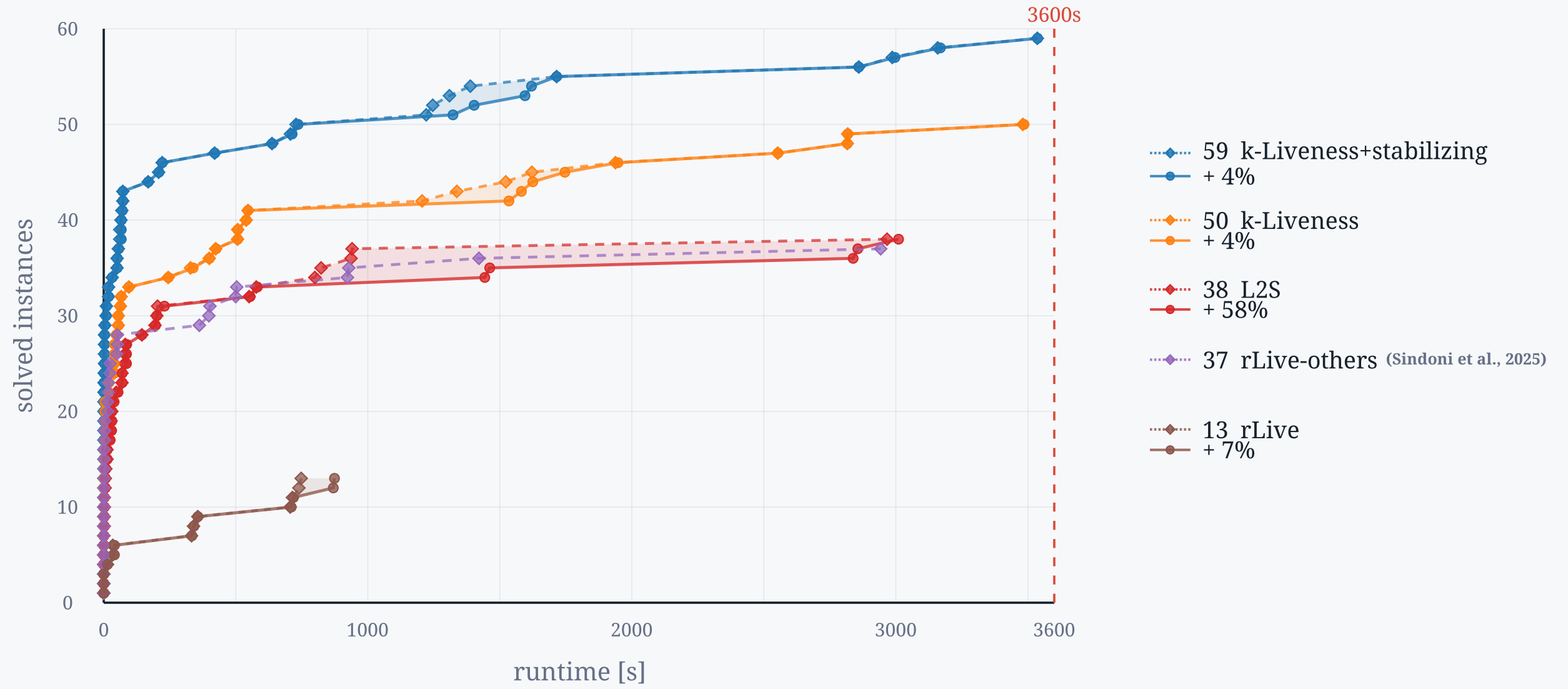




# HWMCC25



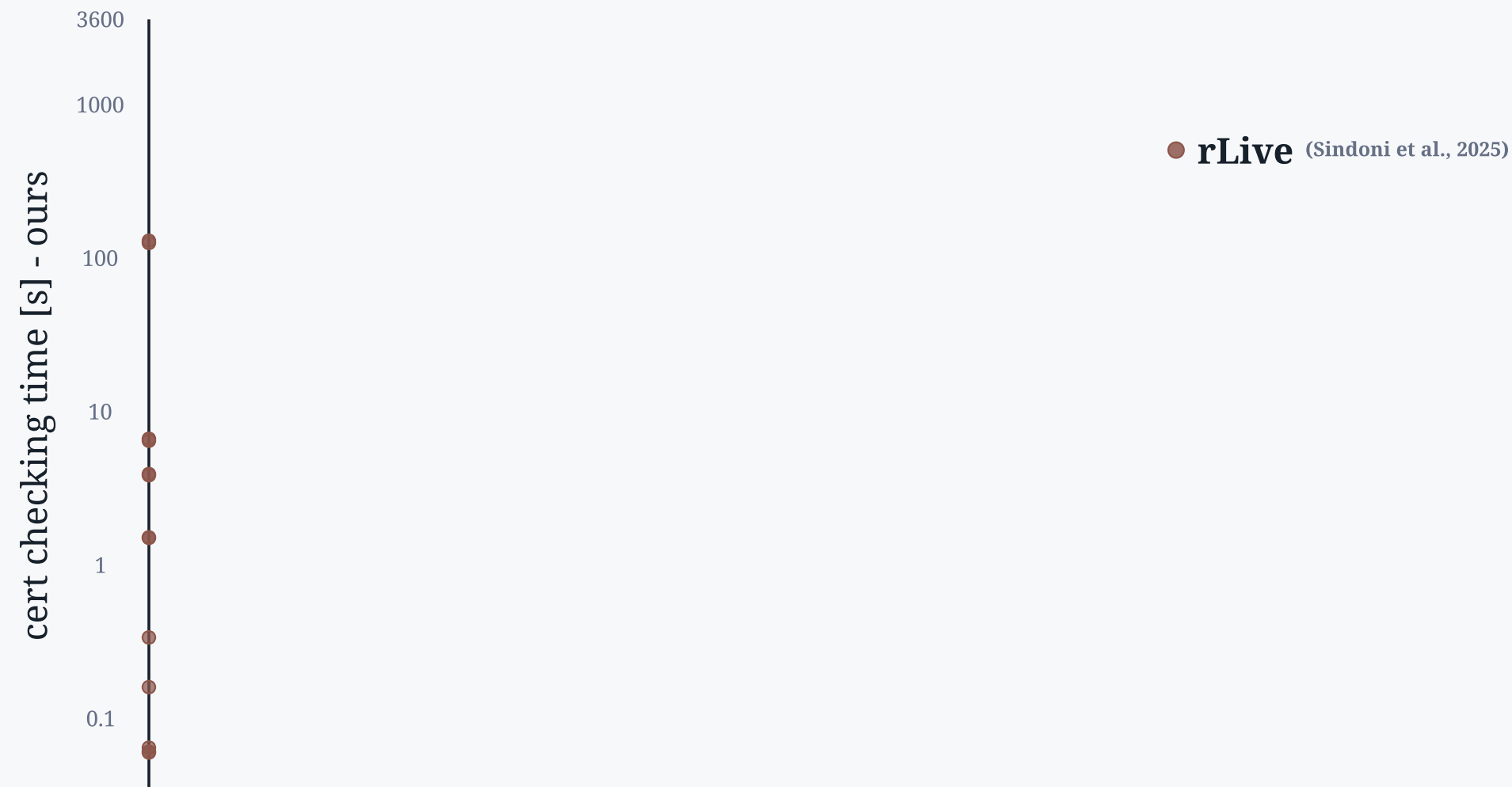
# HWMCC25



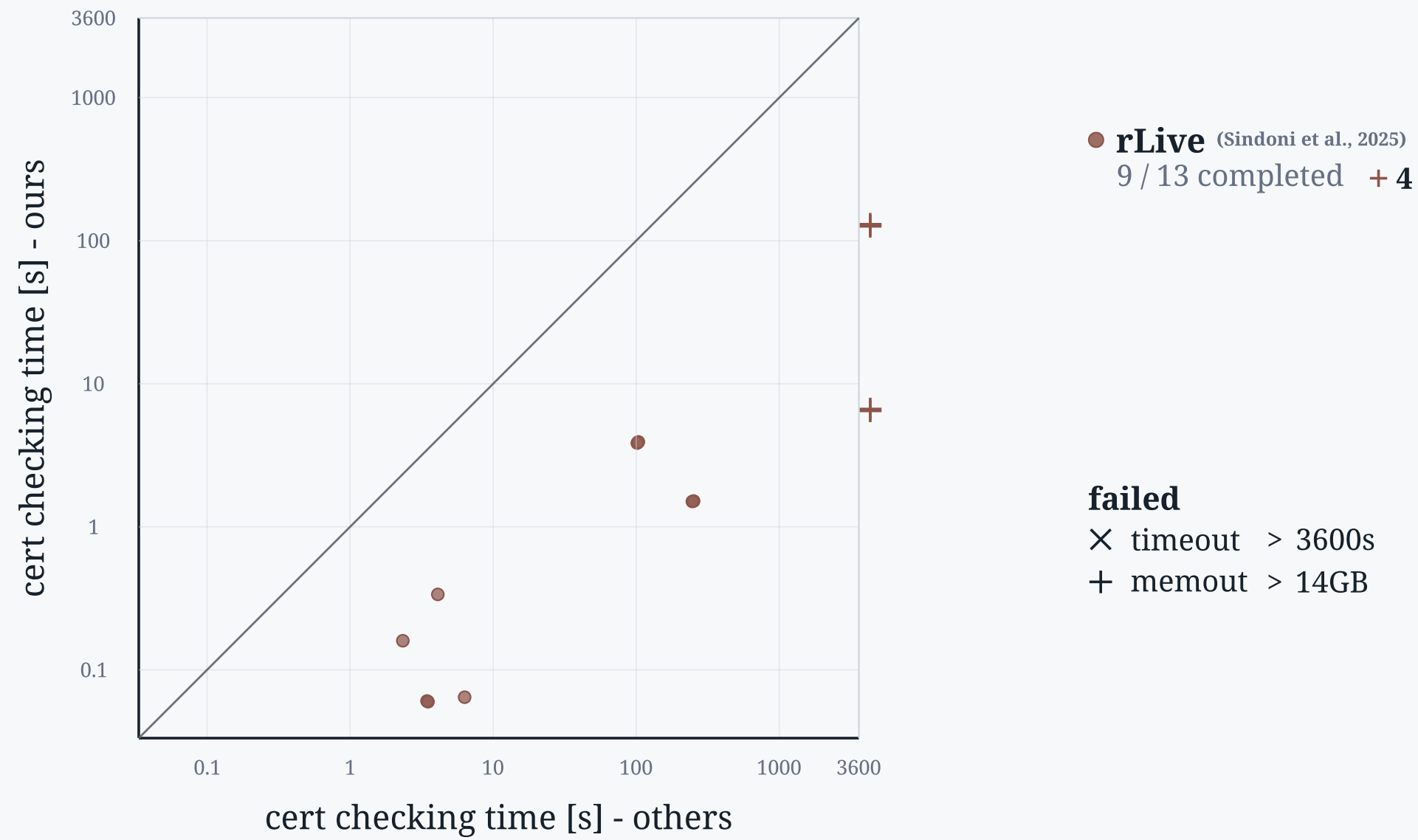
# HWMCC25



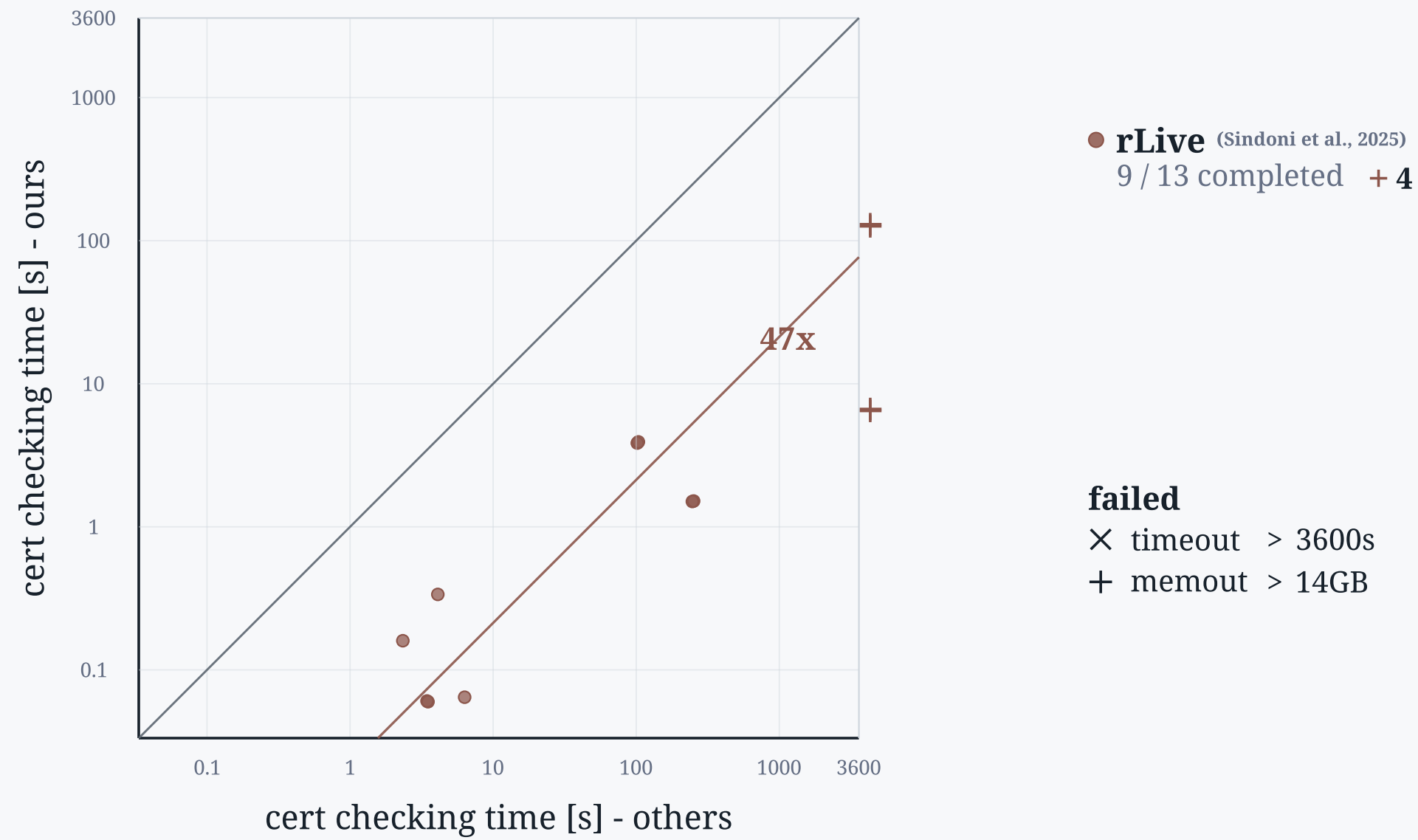
# Certificate checking



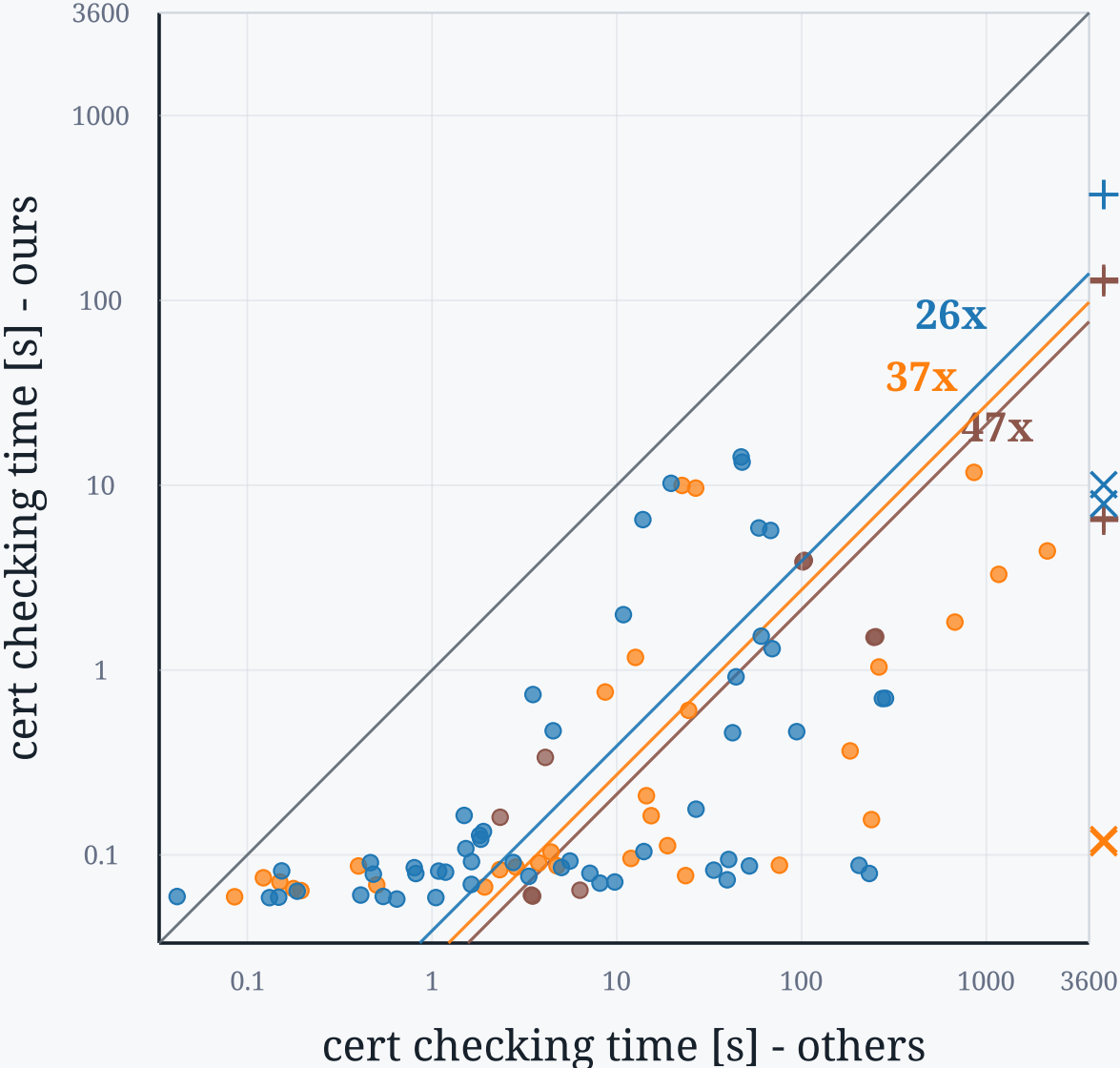
# Certificate checking



# Certificate checking



# Certificate checking



- **rLive** (Sindoni et al., 2025)  
9 / 13 completed + 4
- **k-Liveness** (Griggio et al., 2018)  
31 / 33 completed x 2
- **stabilizing+k-Liveness** (Griggio et al., 2021)  
53 / 57 completed x 2 + 2

**failed**  
x timeout > 3600s  
+ memout > 14GB

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

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**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

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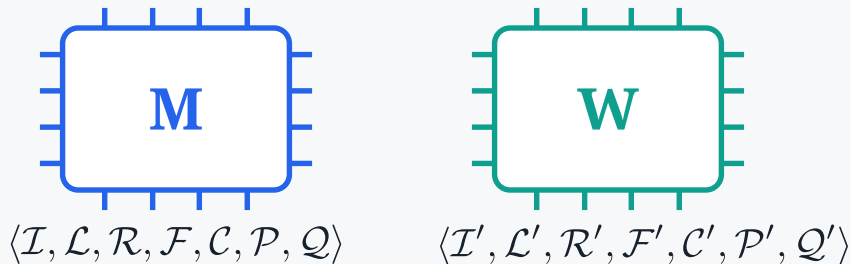
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- [\(Griggio et al., 2021\)](#) Griggio, Roveri & Tonetta, Certifying Proofs for SAT-Based Model Checking, FMSD 2021.
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# Certificate Checking



simulates



inductive

ranked

Reset

$$\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$$

Transition

$$F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$$

Safety

$$C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$$

Liveness

$$\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$$

Base

$$\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$$

Induction

$$F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$$

Decrease

$$\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$$

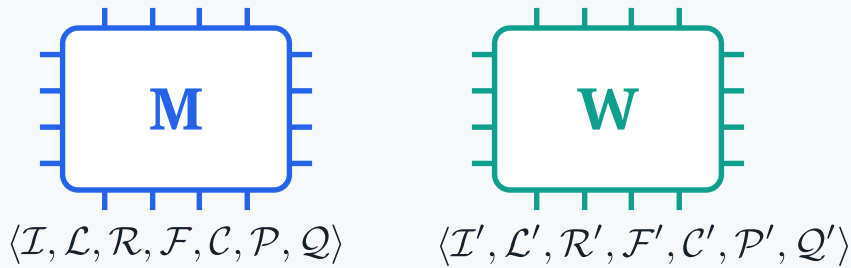
Closure

$$\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$$

SAT Solver

8 checks

# Future work



simulates



inductive

ranked

multiple liveness signals

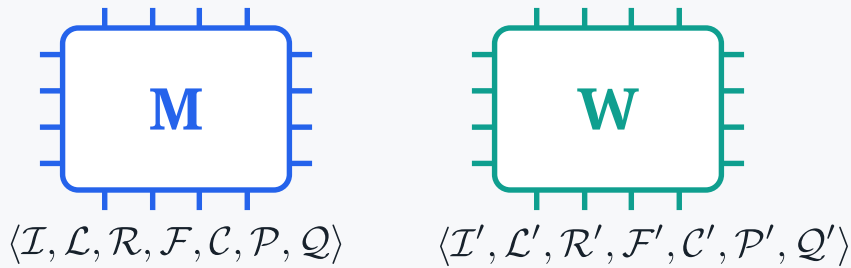
$$\mathcal{Q} = \{q_1, \dots, q_n\}, \quad Q'^{\vee} = \bigvee_{q \in \mathcal{Q}} q'$$

- Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$
- Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$
- Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$
- Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{st} \Rightarrow Q_{st}$
- Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$
- Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$
- Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$
- Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$

SAT Solver

8 checks

# Future work



simulates



inductive

ranked

multiple liveness signals  
 $\mathcal{Q} = \{q_1, \dots, q_n\}, \quad Q'^{\vee} = \bigvee_{q \in \mathcal{Q}} q'$

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow \bigwedge_{q \in \mathcal{Q}} (q'_{st} \Rightarrow q_{st})$

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'_{ts}$

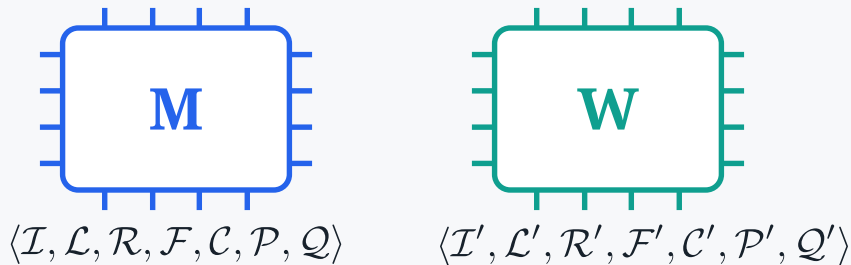
**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'_{su} \Rightarrow Q'_{tu}$



SAT Solver

8 checks

# Future work



simulates



inductive

ranked

multiple liveness signals

$$\mathcal{Q} = \{q_1, \dots, q_n\}, \quad Q'^{\vee} = \bigvee_{q \in \mathcal{Q}} q'$$

Reset

$\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

Transition

$F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

Safety

$C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

Liveness

$\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow \bigwedge_{q \in \mathcal{Q}} (q'_{st} \Rightarrow q_{st})$

Base

$\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

Induction

$F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

Decrease

$\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'^{\vee}_{ts}$

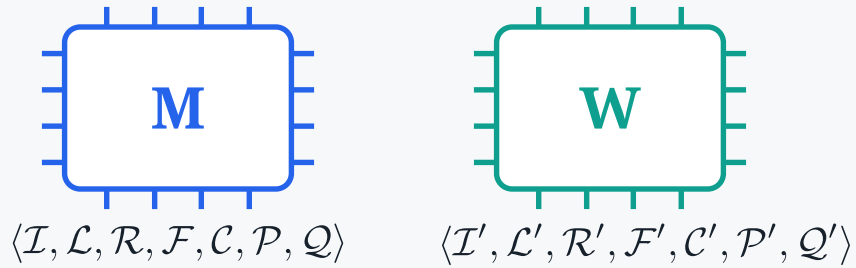
Closure

$\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'^{\vee}_{su} \Rightarrow Q'^{\vee}_{tu}$

SAT Solver

8 checks

# Future work



simulates



inductive

ranked

multiple liveness signals

$$\mathcal{Q} = \{q_1, \dots, q_n\}, \quad Q'^{\vee} = \bigvee_{q \in \mathcal{Q}} q'$$

**Reset**  $\mathcal{R}_s[L \cap L'] \wedge C_s \Rightarrow \mathcal{R}'_s[L \cap L'] \wedge C'_s$

**Transition**  $F_{st}[L \cap L'] \wedge C_s \wedge C_t \wedge C'_s \Rightarrow F'_{st}[L \cap L'] \wedge C'_t$

**Safety**  $C_s \wedge C'_s \wedge P'_s \Rightarrow P_s$

**Liveness**  $\bigwedge_{x \in \{s, t\}} (C_x \wedge C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow \bigwedge_{q \in \mathcal{Q}} (q'_{st} \Rightarrow q_{st})$

**Base**  $\mathcal{R}'_s[L'] \wedge C'_s \Rightarrow P'_s$

**Induction**  $F'_{st}[L'] \wedge C'_s \wedge C'_t \wedge P'_s \Rightarrow P'_t$

**Decrease**  $\bigwedge_{x \in \{s, t\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \Rightarrow Q'^{\vee}_{ts}$

**Closure**  $\bigwedge_{x \in \{s, t, u\}} (C'_x \wedge P'_x) \wedge F'_{st}[L'] \wedge Q'^{\vee}_{su} \Rightarrow Q'^{\vee}_{tu}$

**Consistent**  $\bigwedge_{x \in \{s, t, u\}} (C_x \wedge P_x) \wedge F_{st}[L] \wedge F_{tu}[L] \wedge Q^{\vee}_{st} \wedge Q^{\vee}_{tu} \Rightarrow \bigwedge_{q \in \mathcal{Q}} (q_{st} \Rightarrow q_{tu})$

SAT Solver

9 checks